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Reactive Autonomous Ad Hoc Self-Organization of Homogeneous Teams of Unmanned Surface Vehicles for Sweep Coverage of a Passageway with an Obstacle Course

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Abstract: A team of unmanned surface vehicles (USVs) travels with a bounded speed in an unknown corridor-like scene containing obstacles. USVs should line up at the right angle with the corridor and evenly spread themselves out to form a densest barrier across the corridor, and this barrier should move along the corridor with a given speed. Collisions between the USVs and the corridor walls, other obstacles, and among themselves must be avoided. In the fractions of the scene containing obstacles, the line formation should be preserved, but the demand for an even distribution is inevitably relaxed. This evenness should be automatically restored after such a fraction is fully traversed. Any USV is aware of the corridor direction and measures the relative coordinates of the objects that lie within a given finite sensing distance. USVs do not know the corridor's width and the team's size, cannot distinguish between the team-mates and fill different roles, and do not use communication devices. A computationally cheap control law is presented that attains the posed objectives when being individually run at every USV. The robustness of this law to losses of teammates and admissions of newcomers is justified. Its performance is demonstrated by mathematically rigorous non-local convergence results, computer simulation tests, and experiments with real robots.

Keywords: unmanned surface vehicles; multi-agent; distributed control law; autonomous robots; sensory based robot control



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1. Introduction

Recently, formation control has advanced in a mature stage [1], with an ever increasing attention to the distributed control paradigm and the issue of sensorial and communication constraints. This shift of focus brings strong challenges, both algorithmic and analytical, and makes designs of control laws with rigorously justified non-local convergence a really difficult task [2].

The paper focuses on methods for the spatial self-arrangement of multiple silent mobile robots that are incapable of discerning among their peers, obtain limited sensor data on close objects, and cannot enact different roles and thus have nothing to do but individually run the same algorithm. The issue is, in fact, the generation of effective, coordinated, and cooperative action of the entire team from the separate, homogeneous,

and elementary decision-making of each of its ill-informed members. In computer science, this goes hand in hand with the concept of behavior-oriented artificial intelligence [3].

The above multifaceted uniformity entails several benefits regarding cost, scalability, fault tolerance, flexibility, repair, and implementation, and is among the key topics studied under the umbrella of swarm robotics [4]. Meanwhile, that uniformity is problematically compatible with many popular approaches to decentralized formation-building control [2]. Among them, there are, e.g., methods using virtual rigid or flexible structures, either team-wide or local leaders/beacons, or designated data or command links between specially selected agents, as well as the assignment of diverse entities, like labels, tasks, goals, etc., to different team members.

The uniformity previously characterized is better consistent with behavior-based approaches [5], such as virtual forces or artificial potential methods. These often lack sufficient guarantees that the targeted formation does emerge, although exceptions are available among distributed consensus protocols [6,7] and related algorithms for formation control [8] or coverage [9] control.

This paper presents a distributed algorithm for the self-deployment of fully homogeneous robotic teams for efficient sweep coverage of an area [10]. The field of coverage control is becoming progressively important due to the growing application of robotic teams to monitoring or processing extended objects, either natural or human-made, and vast areas. In the meantime, the development of microelectronic technologies gives the practical possibility to use large-scale networks of inexpensive devices for executing such missions. Their examples include [11] automatic mine-sweeping, reconnaissance, maintenance inspection, search and rescue operations, finding the source of a pollutant, the monitoring of bush fires, environmental extremum seeking, and target seeking, to name just a few.

To effectively carry out such coverage missions, the team members should be put in appropriate, ideally optimal, locations; at a minimum, any point of the handled region should be shadowed by some member. For large teams, this may be a burdensome task when being performed by external means. Hence, the self-deployment of mobile elements is an appealing option: all and everyone should arrive at the targeted deployment autonomously and on their own. To this end, distributed navigation strategies are needed that merge the easiness of implementation, computational cheapness, and low consumption of resources with guarantees of convergence. Also, the strategy should be robust to failures and the additions of the robots, and should entrust all of them with identical roles, thus making them fully interchangeable. The satisfaction of these partly competing demands constitutes a real challenge in practice.

The paper treats the problem of not static, but dynamic coverage: this problem integrates the sweep and barrier coverage patterns [12]. Effective coverage of a priori unknown scene with a robotic team of an unknown size is handled. The coverage is troubled by unknown obstacles in the area, and so the appropriate positions of the robots in the scene cannot be pre-computed, but are to be built on-the-fly in a feedback fashion based on only very limited sensory information. The mission objectives include lining up the team across an unknown passageway (either natural or human-made, either physical or virtual), frontally moving the thus obtained straight row of the robots with a given speed, preserving the row formation even if the row is cut into pieces with an obstacle, and automatic self-distribution of the robots over the width of the current passage into the densest possible net-like barrier. (Similar requirements may arise when a regular linear formation of mobile robots has to go through a confined environment cluttered with obstacles.) All these should be achieved by a computationally cheap control rule that is common for all robots and is to be individually run on every of them.

This sort of coverage is important for various applications, including mine sweeping, the deployment of static sensor networks, ecological surveillance, sentry duty, rescue operations, and maintenance inspection, to name just a few. However, among the survey of papers on multi-robot self-organization for the sweep coverage of corridor-type scenes, the authors failed to come across one treating this issue in the presence of obstacles, in the face of the above uncertainties and limitations, and with the goal to keep a straight row formation and densest distribution over an unknown passage despite the obstacles. This paper partly fills these gaps.

Specifically, it handles the case of a polygonal corridor environment with finitely many polygonal obstacles, which are a priori unknown. There is a fully homogeneous team of speed-limited USVs, each modeled as a continuous-time integrator. Every USV senses the scene within a given finite distance and thus determines the relative positions of objects that are in direct line of sight. The USVs do not know the width of the corridor, cannot tell the difference between any two peers, cannot fill distinct roles, and do not use communication devices. The above “polygonal obstacles” may in fact be upper approximations of obstacles of more general nature with polygons.

The principal contributions of this paper are as follows. (A) The problem is solved for a fully homogeneous team of robots (in particular, without the shared numbering of them) and, moreover, in a practical situation where any robot has access to data about a peer not only if the latter is within a given “visibility range” but also if nothing obscures the view to it. (B) The proposed control rule is computationally inexpensive, partly since it is reactive: in real time, it directly converts the current observation into the control input in a reflex-like fashion. (C) The problem is solved with respect to an unknown corridor-like environment cluttered with unknown obstacles. (D) The straight-line cross-section formation of USVs and their succession in the formation are maintained even if the line formation is broken into mutually invisible pieces by obstacles. Furthermore, even spacing is automatically restored after the obstacle course is left behind. (E) The proposed control law excludes the collisions of the USVs with one another, the obstacles, and the corridor walls. (F) The situation of both losses and the additions of team members is elaborated; it is proven that, under the proposed control rule, the team automatically adapts to changes in its composition. (G) Mathematically rigorous guarantees of convergence and justification of the performance are provided. (H) The rule is proven to succeed even if there is no informational connectivity of the team at the start of the mission since it creates connectivity from nothing sooner or later. Also, the algorithm restores this connectivity after it happens to be broken due to either the dropouts of team members or the visibility-blocking effect from obstacles. (I) It is shown that whenever the tasks of even distribution across the corridor and obstacle avoidance are to be solved simultaneously, the complexity of the control system can be reduced by omitting (unlike the standard approach) a special obstacle avoidance module since its functions are automatically performed by the control module responsible for distributing the robots across the corridor. (J) Unlike most works dealing with corridor environments, we handle corridors with non-straight, polygonal walls. (K) The presented findings are based on the application of a complete set of general research techniques that include theoretical developments, computer simulation tests, and real-life experiments.

The property (B) is among the main targets of the reported design, bearing in mind applications to small and miniature robots, as well as the reservation of computational resources for basic, higher-level objectives of the mission. Whereas (I) clearly participates in (B), it is worth noting that the property (I) holds under certain circumstances (detailed in the body of the paper).

Regarding (H), we remark that, for the bulk of studies on distributed formation building methods based on local data, informational connectivity is a typical prerequisite for convergence; see, e.g., [6,13,14]. It is often merely postulated that this connectivity persists over the infinite time horizon in one form or another due to an unidentified reason. However, if data transfer is distance-dependent, execution of the main motion control law may endanger this assumption. In such a situation, some algorithms rely on initial connectivity and block the robots' moves that cause the breakages of any existing data links; see, e.g., [15–17]. However, then the control goal may be unattainable (e.g., if the robots start in a tight cluster and so enjoy all-to-all visibility but have to spread over so vast an area that the distance between some robots inevitably exceeds the visibility range). Another approach preserves connectivity, while allowing the breakages of cautiously selected links [18–20]. The price for this is high computational and communication burdens, and the need for awareness of the global topology of data exchange. By contrast, the proposed algorithm creates informational connectivity from nothing, and does so at a minimal computational cost and based on only local data.

The body of this paper is organized as follows. Section 2 offers an overview of prior relevant research, whereas Section 3 states the problem. Section 4 presents the robots model and assumptions. Sections 5 and 6 describe the preliminary stage of data processing and the proposed navigation law, respectively. The main theoretical results are given in Section 7. Sections 8 and 9 report on the results of computer simulations and real-world tests, respectively. Section 10 offers conclusions. The proofs of the key results are given in Appendix A.

2. Related Work

The bulk of the literature on robotic coverage algorithms is devoted to path planning: how to build a path that passes close to any point from the area of interest; we refer the reader to [21–23] for recent surveys of this field. Also, an extensive body of research is concerned with the nodes' placement and dispatch problems [24–29], where the main concern is to find out where, how many, and how nodes should be distributed. By and large, these lines of research focus on geometrical aspects of the problem, leaving the issues of self-spreading and motion control beyond elaboration.

There is also a body of research on reactive algorithms for robots' self-distribution in coverage missions by using some sort of feedback control [23,30]. Some of these methods are stochastic, often attempt to copycat natural processes in biology, and are surpassed by the idea of the swarm intelligence [31]; see, e.g., [32,33] for representative samples. Another group consists of deterministic physicomimetic methods and draws much inspiration from the concepts of virtual forces, artificial potentials, and molecular equilibrium phenomena [23,30,34–37]. The third category is distinguished by its reliance on concepts and techniques of geometry [23,30,38]. The Voronoi tessellation is a popular tool here [29,39–43]. Another approach is based on the access of all robots to a common global geometric structure, like a regular grid ([9], Ch. 6) or cell decomposition [44].

For more details on the conceptual and taxonomic aspects of the area, we refer the reader to the specialized surveys, e.g., [23,30,38,45]. Overall, it mainly focuses on achieving the blanket or barrier coverage of a given structure [12], the objectives of which are critically different from forming an evenly spaced straight line formation moving in an unknown corridor. Many approaches are not at all aimed to build a regular formation, need a rather comprehensive knowledge of the environment or some pre-specified global structure (which may call for access to GPS), and necessitate a heavy computational burden. Also, rigorous guarantees of convergence are relatively rare, and the algorithm validation often comes to simulation tests and secondary theoretical facts.

Challenges of the robot's autonomous navigation in corridor-like scenes have received substantial attention from the robotic community due to their own value and since they are common constituents of more complex environments; see, e.g., [46–51] and the literature therein. However, the respective literature largely deals with the case of a single robot, though exceptions also exist; see, e.g., [52–55]. In [52,54], an algorithm for the sweep coverage of a corridor with straight walls is presented and rigorously justified, postulating persistent informational connectivity of the team. Corridors with arbitrarily shaped walls are handled in [55], where a shared numbering of the teammates and steady designated information links between some of them are assumed (i.e., the team is not fully homogeneous), and forming a straight-line formation is not an objective. The homogeneity of the team is also violated in [53], as the algorithm employs role hierarchy among the robots. In all these works, corridors without obstacles were considered; in particular, the loss of sensorial contact and the breakage of informational connectivity because of obstacles were not addressed.

For navigation in the obstacle-free part of the corridor, we use some findings from [56]. However, they are critically insufficient to deal with obstacles, and the main focus of this paper is on the design and elaboration of respective additions to the algorithmic technology. Relative to a considerably simplified version of the scenario from this paper, some preliminary algorithmic ideas are reported (without proofs) in the conference publication [57]. As compared with it, the current text offers significant extensions regarding both the scenario and algorithm (which in fact essentially differs from that from [57]), as well as a fully renewed conceptualization of the both. In addition, this manuscript offers a complete and rigorous theoretical justification of the proposed algorithm and obtained results, discusses a whole number of extra issues (like robustness against changes in the team membership, etc.), and not only presents the results of new computer simulation studies but also logically completes the research with real-world experiments.

3. Sweep Coverage of a Corridor

We consider N point-wise planar USVs, enumerated by $i = 1, \dots, N$, which travel in an unknown planar *corridor* $\mathcal{C}$ between two parallel straight lines. (Later on, corridors with non-straight walls will be addressed.) The USVs should ultimately go along the corridor with a common pre-specified speed $v_{\rightarrow} > 0$. Furthermore, they should find themselves on a common moving cross-section $\mathfrak{N}$ oriented perpendicularly to the walls; see Figure 1. And lastly, they should evenly distribute themselves over $\mathfrak{N}$ to form the densest net across the corridor.

The mission is troubled by unknown obstacles $O_1, \dots, O_K$ inside the corridor (see Figure 1). The USVs must not collide with the corridor walls, the obstacles, and one another. Hence, the even distribution across the entire width of the corridor is possible only in its obstacle-free part: even if the team approaches the obstacle course in the ideal order, this order is then inevitably violated. The navigation law must automatically restore that order after the obstacles have been left behind.

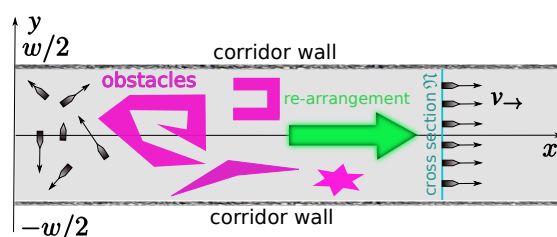


Figure 1. Sweep coverage of a corridor with an obstacle course.

We consider point-wise USVs with simple integrator kinematics and a given speed bound $\bar{v}$:

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \quad \|\mathbf{v}_i\| \leq \bar{v}. \quad (1)$$

The velocity vector $\mathbf{v}_i = \mathbf{v}_i(t) \in \mathbb{R}^2$ is the control input. Every USV has its own local frame of reference. In this frame, it identifies the corridor orientation and the relative coordinates of the objects, including pieces of the obstacles and the walls of the corridor, if they are within a given finite *visibility distance* $R_{\text{vis}} > 0$ and the line of sight to them is not blocked. We assume that no USV can fully block the sight of the peers at any obstacle due to some reason, e.g., the superiority of the obstacles over the USVs in size or since USVs can “see” the obstacles at a height above the peers.

None of the USVs use communication devices and know the number N of team members or the width w of the corridor. The team is fully homogeneous: the USVs cannot tell the peers apart, there is no hierarchy among them, and they are unable to play distinct roles and must therefore be driven by a common control rule. So, the posed tasks are to be solved in an autonomous and distributed fashion.

Formal Problem Statement

Unlike the USVs, we can use a right-handed *absolute* frame; its x axis goes in the targeted direction of motion parallel to the corridor walls in the middle of them; see Figure 1. Thus, the corridor walls are described by the equations $y = \pm w/2$. For the convenience of exposition, the x and y directions are said to be *horizontal* and *vertical*, respectively.

The location of USV i at time t is denoted by $\mathbf{r}_i(t) = [x_i(t), y_i(t)]$, and the polar angles are counted counterclockwise from the x axis.

To specify the problem, we introduce the symbol $\mathbf{p}_j(a)$ for the point $\mathbf{p}$ with the abscissa $x(\mathbf{p}) = a$ and the ordinate $y(\mathbf{p}) = -w/2 + jw/(N+1)$. It is easy to see that the points $\mathbf{p}_1(x), \dots, \mathbf{p}_N(x)$ evenly partition the common cross-section of the corridor described by the equation $x = a$. The pursued navigation objective is in fact much disclosed in the following.

Definition 1. *The USVs are said to sweep the scene with a speed of $v_{\rightarrow}$ if the following holds true:*

- (i) *Any USV i does not collide with the obstacles $\mathbf{r}_i(t) \notin O_k \forall k, t$, the walls of the corridor $|y_i(t)| < w/2 \forall t$, and the other USVs $\mathbf{r}_i(t) \neq \mathbf{r}_{i'}(t) \forall t, i' \neq i$;*
- (ii) *There is x_* and an enumeration of the USVs such that $\|\mathbf{r}_i(t) - \mathbf{p}_i(v_{\rightarrow} \cdot t + x_*)\| \xrightarrow{t \rightarrow \infty} 0 \forall i$.*

It is needed to design a control rule that is common for all USVs and, when being independently run on every one of them, ensures the property from Definition 1, starting from occasional, though more or less reasonable, initial positions.

4. Theoretical Assumptions About the Scenario

To introduce and even more to discuss the proposed navigation algorithm, we need to first highlight some assumed properties of the scene. To streamline the exposition, we embed this material into a complete description of the assumptions underlying our theoretical analysis.

Assumption 1. *The obstacles are disjoint polygons (not necessarily convex) and lie inside the corridor.*

This does not impose significant limitations on the obstacles since first, what is handled as an “obstacle” in robotics research, is often an (upper) approximation of a real obstacle with polygonal areas and second, this approximation can be made as accurate as one wishes.

To proceed, some definitions are required. A ray (segment of a line) is said to be *open* if it does not contain its root (its end-points). An *upward/downward* ray is a vertical ray that propagates upwards/downwards. A set is said to be O_k -free if it has no points in common with the obstacle O_k .

Definition 2. A upward/downward ray $\mathcal{R}$ is said to be forward critical for O_k if the following holds:

- (i) It is open and goes from the boundary ∂O_k of the obstacle O_k ;
- (ii) This ray is not O_k -free;
- (iii) To the left of $\mathcal{R}$ and arbitrarily close to it, there is an upward/downward ray that is O_k -free.

In Figure 2, the green rays are not O -free and are not critical, the red ones are critical, and the blue rays are O -free and are not critical. Critical rays mark either forward vertical sides of the obstacle or entrances to its “deadlock caves” (a *cave* of O is defined as a connected component of the set of points p such that the rays issued from p both upwards and downwards, respectively, intersect O). The concepts of such a ray and cave are born from the idea to bypass the obstacle by means of constantly and simultaneously (1) following its boundary and (2) progressing down the corridor $\dot{x} > 0$. After arrival at a front vertical side, executing (2) would imply penetration into the obstacle and so is impossible. The attribute “deadlock” marks the caves such that the robots may enter but cannot leave them while following the instructions (1) and (2). Figure 2a illustrates a deadlock cave, which is entered while following the pink side of the obstacle and then cannot be left without violating (2). The caves in Figure 2b,c do not do any harm in this regard since they are not entered due to (2). When arriving at their entry points, like A in Figure 2b,c, the rule (1) brings about an instantaneous jump to point B and subsequently following another piece of the boundary to the right.

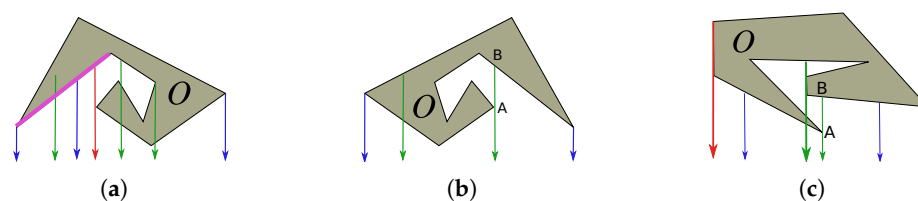


Figure 2. Critical (red) and non-critical (other colors) downward rays. (a, c) Obstacle with critical (red) ray; (b) Obstacle without critical (red) rays.

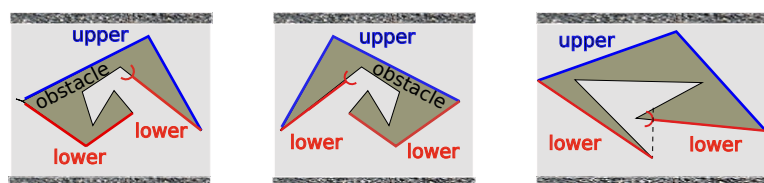


Figure 3. Lower and upper fences.

Assumption 2. No obstacle has forward critical rays.

Example 1. Any convex obstacle with a unique leftmost point meets Assumption 2.

Example 2. We orient the boundary ∂O of an obstacle $O := O_j$ so that O is to the left when traveling over ∂O , and consider the cycle C of the consecutively adjacent edges forming ∂O . Let C be cut into two (non-cyclic) chains so that the edges from a common chain are consecutively concatenated, the polar angles of the edges from one of the chains are acute (less than $\frac{\pi}{2}$ in absolute value), and obtuse for the other chain. Then, Assumption 2 is true.

Assumption 2 rejects the obstacles like those in Figure 2a,c. Some of them can yet be put into the framework of Assumption 2 if every robot appropriately modifies its interpretation of the scene on the fly. Some hints of such a kind will be discussed in Section 7.8.

To state the last assumptions, a few more definitions are useful.

Definition 3. A point p of the obstacle's boundary ∂O_k is said to be (globally) lower/upper if p is the upper/lower end of a vertical open O_k -free (ray) segment. The closure of any set of such points is said to be (globally) lower/upper.

Remark 1. An lower/upper side of the polygon O_k cannot be vertical. Conversely, any non-vertical side of the polygon O_k is either lower or upper.

The union of all globally lower/upper sides of the obstacle O_k is called its lower/upper fence $\mathcal{F}_k^\mp$; see Figure 3.

It is clear that the abscissas of different points of $\mathcal{F}_k^\pm$ differ. We orient the lower/upper fence and any its segment in the direction of the abscissa ascending. Then, the polar angle of any such segment is acute. We treat the “upper” $\mathcal{E}^+ := \{p : y \geq w/2\}$ and “lower” $\mathcal{E}^- := \{p : y \leq -w/2\}$ exterior of the corridor as obstacles with positively oriented boundaries. The fences of these obstacles are equal to the respective walls of the corridor.

Our final assumptions are concerned with the mission feasibility. First, the feedback from the distance gaps between the robots should not be lost under the targeted even deployment of the team across the corridor, which yields that $\frac{w}{N+1} \leq R_{\text{vis}}$. Second, the requested speed $v_{\rightarrow}$ should match the capacity of the robots: $v_{\rightarrow} \leq \bar{v}$. Moreover, the tactics of the boundary following may involve tracking parts of the upper or lower fences of obstacles. When doing so, the top speed of the robot's abscissa is defined by the polar angle α of the currently tracked segment and is equal to $\bar{v} |\cos \alpha|$. So, the demand for the value $v_{\rightarrow}$ of the horizontal speed is reasonable only if $v_{\rightarrow} \leq \bar{v} |\cos \alpha|$.

By slightly enhancing these necessary conditions via $\leq \mapsto <$ and introducing an upper bound $\bar{\alpha} \in (0, \frac{\pi}{2})$ on the absolute values of the polar angles of the segments in all upper and lower fences, we arrive at our final assumption:

$$v_{\rightarrow} < \bar{v} \cos \bar{\alpha}, \quad w/(N+1) < R_{\text{vis}}. \quad (2)$$

5. Navigation Algorithm: Preprocessing Sensory Data

This section opens with a description of the navigation algorithm by focusing on the employed technical geometrical computations. Every robot performs them individually and from its own measurements: it imaginarily modifies the visible parts of the obstacles and computes their features.

Endowing the obstacles with visors aims to avert an unlikely, though not impossible, head-on collision with the front vertex $p_{\min}^k$ of an obstacle O_k (like in Figure 4a). To this end, the USV imaginarily affixes a short horizontal visor to $p_{\min}^k$, as is shown in Figure 4b, and uses the so-obtained visored obstacle when computing the control input.

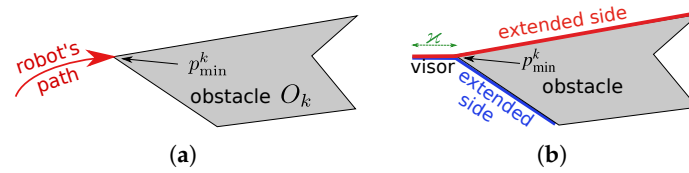


Figure 4. Artificial visor of an obstacle: (a) Unlikely situation; and (b) Artificial visor.

The rationale behind this hint stems from the wish to reduce the complexity of the control software and hardware by omitting specialized obstacle avoidance components. The idea is to show that an “inevitable” control module designed for distributing the robots across the corridor is able to largely guarantee collision avoidance by its own right. Because of its main concern, this module basically explores the scene vertically. Adding the visor entails that the USV starts reacting to any obstacle before actually reaching the area of its vertical shadows. By applying the routine machinery of receding from the vertically close peers (which is a necessary part of the above module) to the obstacle points (both true and imaginary), the USV finds itself either below or above the obstacle when that area is actually reached. As a result, head-on collisions are avoided.

Now, we delve into the details. For any USV, the *skeleton* S is the set of the currently visible pieces of the obstacles’ boundaries. A *visible front cusp* is a part of S that lies down the corridor with respect to the USV and looks like either of the following:

- (A) Two closed segments that go from a common point p down the corridor (see Figure 5a);
- (B) A single closed segment with an end-point p such that the USV “sees” no continuous extension of S to the other side of p (see the red part of Figure 5b).

The case (B) holds if the obstacle blocks the line of sight to one of its own sides adjacent to p , unlike the blue part of Figure 5b, where the block is due to another obstacle (this can be recognized by an instantaneous drop of the distance to the visible side along a ray which continuously rotates about the robot so that the point of reflection slides towards p). In Figure 5b, the visible blue horizontal segment is not closed since point p_* , being invisible, does not belong to it. Hence, this part is not a visible front cusp by definition. By the same reason, the “downstream” visible part from Figure 5c is not a visible front cusp since point p_* does not belong to it: this point does not lie “downstream” of the USV and so belongs to neither the skeleton nor any its parts.

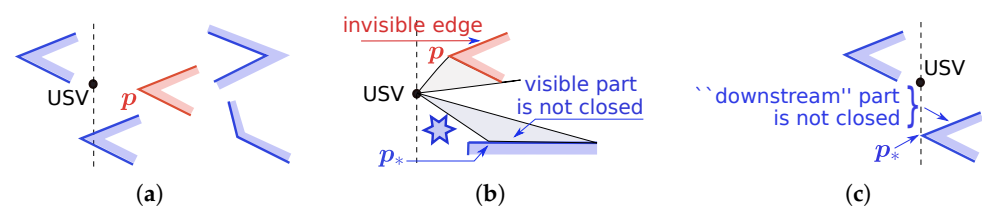


Figure 5. Angles that are (red) and are not (blue) visible front cusps. (a) The case (A) of visible front cusps; (b) The case (B) of visible front cusps and the example of not closed visible part; (c) The example of not closed “downstream” visible part.

Whenever the USV finds cusps, it adds a straight horizontal visor V with a length of $\varkappa$ to every of them, as illustrated in Figure 4b. Here, $\varkappa > 0$ is a free parameter. If the USV finds itself exactly on the visor V , it yet adjudges (at random) that it is either “above” or “below” it. In spite of everything, the USV sticks to this judgment while V remains visible: even if the USV happens to get to the other side of V , it vertically displaces the visor in its “mind” so to put itself at the displaced visor and then treats the scene accordingly. Then, the USV does not “see” the peers that are imaginarily “on the other side” of the straight

line hosting the visor. Meanwhile, it may turn out that such a peer treats the USV at hands as “visible”. The points of the visors are simultaneously lower and upper, and so the visors become parts of both lower and upper fences.

The outcome of these speculative manipulations with the visor will be commented on in Remark 3.

Further geometrical definitions and preliminaries. Let E be a compact subset of the corridor. A point p is said to be *in the x -scope of E* if this point is vertically shadowed by E :

$$\min_{r \in E} x(r) \leq x(p) \leq \max_{r \in E} x(r).$$

If, in addition $p \notin E$, there is a vertical straight segment (p, r) such that $r \in E$ and $(p, r) \cap E = \emptyset$. If this segment is above the point r , the point p is said to lie *above* E , and *below* it otherwise. Any point in the x scope of E lies either above, or below, or simultaneously above and below E .

Points p_1 and p_2 are said to be *separated by the obstacle O_k* if $p_i \notin O_k$ and $(p_1, p_2) \cap O_k \neq \emptyset$. For robot i and time t , the *laterally visible obstacle point* is a point p from the union $O_{\cup} := \bigcup_k O_k$ of the obstacles such that the segment $(p, r_i(t))$ is vertical, does not intersect $O_{\cup}$, and is less than R_{vis} in length. There is no more than one (and, maybe, none) such point both above and below the USV.

Procedure of preprocessing sensory data. At any time t , any USV i builds the following.

The set $\mathcal{N}_i(t)$ of visible peers: it consists of i and all peers j such that $\|r_j(t) - r_i(t)\| \leq R_{\text{vis}}$ and the view of j is blocked by no obstacle. Since $i \in \mathcal{N}_i(t)$, the number of elements $\#\mathcal{N}_i(t) \geq 1$.

The set $\mathcal{N}_{i,\delta}(t) := \{j \in \mathcal{N}_i(t) : |x_i(t) - x_j(t)| \leq \delta\}$ of close peers. Here, $\delta > 0$ is a parameter of the algorithm.

The set $\mathcal{L}_{i,\delta}(t)$ of essential lateral points that consists of $r_j(t), j \in \mathcal{N}_{i,\delta}(t)$ and the laterally visible obstacle points (if they exist). Let $L_{i,\delta}^{\pm}(t)$ be the set of $p \in \mathcal{L}_{i,\delta}(t)$ such that $\pm[y(p) - y_i(t)] \geq 0$.

The lateral free space is defined by the following formula:

$$d_{i,\delta}^{\pm}(t) := \begin{cases} \min_{(x,y) \in L_{i,\delta}^{\pm}(t)} \pm(y - y_i) & \text{if } L_{i,\delta}^{\pm}(t) \neq \emptyset, \\ R_{\text{vis}} & \text{otherwise.} \end{cases} \quad (3)$$

The index $\pm$ tells us whether the space above $+$ or below $-$ USV i is assessed.

To avert crashes with the obstacles, more scene features are to be taken into account.

Graph $\Gamma_i(t)$ of the close visible peers is based on free (nominally small) parameters $\gamma_x, \gamma_y > 0$. This is the undirected graph with the set of nodes $\mathcal{N}_i(t)$, where j and j' are linked if and only if

- $|x_{j'}(t) - x_j(t)| \leq \gamma_x$ and $|y_{j'}(t) - y_j(t)| \leq \gamma_y$;
- No obstacle separates USVs j and j' ;
- If some of them (say j) lie below/above a visor, then the ordinate of the other USV j' is less/greater than that of this visor provided that USV j' is to the left of USV j .

Straight-line graph $\mathcal{G}_i(t)$ of the intimate peers is the union of the segments $[r_j(t), r_i(t)]$ over j 's from the connected component of $\Gamma_i(t)$ that contains i .

Upper/lower base $B_i^{\pm}(t)$ is the set of points $p \in \mathcal{G}_i(t)$ with the following properties:

- (B.1) Vertically below/above p USV i “identifies” an obstacle O_k at a distance $\leq \gamma_y$;
- (B.2) If $p = r_j(t)$, then the infinitesimal shift of p in the x direction does not increase the vertical distance from p to the obstacles (see Figure 6);

(B.3) If $p \in \mathcal{S} := (r_j(t), r_{j'}(t))$ for a link in $\mathcal{G}_i(t)$, then the infinitesimal plane-parallel shift of $\mathcal{S}$ in the x direction does not increase the vertical distance from $\mathcal{S}$ to the obstacles.

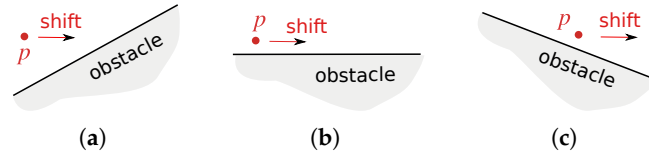


Figure 6. (a,b) A base point p and (c) a non-base point.

USV i is called the *evader* if either there exists a point $p \in B_i^+(t)$ whose ordinate $y(p) \leq y_i(t)$ or there exists a point $p \in B_i^-(t)$ whose ordinate $y(p) \geq y_i(t)$, and only one of these cases occurs. The *main base* is the base addressed in the case that does occur. The concept of evader is introduced to identify the case where USV should take special measures against obstacles since the idea to go straight ahead along the corridor runs the risk that either the USV at hands or its intimate peer may collide with the obstacles, or their intimacy may be destroyed by an obstacle.

Avoidance angle $\alpha_i(t)$ is built so that moving at this angle (to the x axis) creates no threat of collision with the obstacles in the near future; see Figure 7. It is defined based on a parameter $\alpha_* > 0$ (to be specified later on) and the notation $\mathcal{V}$ for the union of all visors:

$$\alpha_i(t) := \begin{cases} \alpha_* \text{ if } \begin{cases} r_i(t) \notin \mathcal{V}, \text{ USV } i \text{ is an evader} \\ \text{and its main base is } B_i^+(t), \\ \text{or the USV is on a visor} \\ \text{and "thinks" to be above it,} \end{cases} \\ -\alpha_* \text{ if } \begin{cases} r_i(t) \notin \mathcal{V}, i \text{ is an evader} \\ \text{and its main base is } B_i^-(t), \\ \text{or the USV is on a visor} \\ \text{and "thinks" to be below it,} \end{cases} \\ 0 \text{ in all other cases.} \end{cases} \quad (4)$$

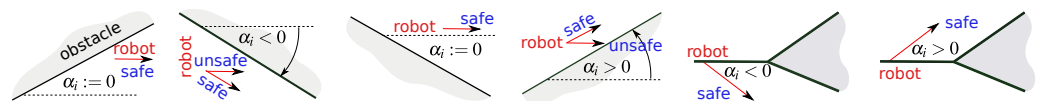


Figure 7. Basic idea behind the avoidance angle $\alpha_i(t)$.

6. Navigation Algorithm

This algorithm uses two real functions $F(\cdot)$ and $G(\cdot)$ with special properties illustrated in Figure 8. Both functions are continuous and piecewise continuously differentiable; they are to be preliminarily selected by the designer of the control system. The function $F(\cdot)$ is defined on $\mathbb{R}$ and is such that:

$$xF(x) > 0 \quad \forall x \neq 0, \quad \mathcal{F}'(0\pm) > 0, \quad \bar{F} := \sup_{x \in \mathbb{R}} |F(x)| < \infty, \quad \sup_{x \in \mathbb{R}} |F'(x\pm)| < \infty. \quad (5)$$

The function $G(\cdot)$ is defined on the interval $[0, 2R_{\text{vis}}]$, is constant to the right of some point R_G of this interval, and is such that

$$G'(L\pm) > 0 \quad \forall L \in [0, R_G], \quad G(L) = \bar{G} \quad \forall L \in [R_G, 2R_{\text{vis}}], \quad G(0) = 0. \quad (6)$$

Here, $R_G \in (0, R_{\text{vis}}]$ and $\bar{G} > 0$ are viewed as controller parameters. For example, the above requirements are met by $F(x) = a\lambda_*(x/b)$, $G(L) := cL + d \forall L \in [0, R_G]$, $G(L) = cR_G + d \forall L \in [R_G, 2R_{\text{vis}}]$, where $a, b, c > 0, d \geq 0$ are parameters and $\lambda_*(z) = \arctan(z)$, $\frac{z}{\sqrt{1+z^2}}$.

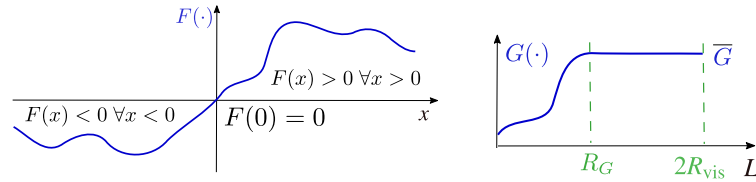


Figure 8. Two functions needed to build the navigation law.

Control rule uses one more free parameter $P > 0$. The control signal v_i (i.e., the velocity vector) for USV i is defined in terms of its abscissa v_i^x and ordinate v_i^y :

$$v_i^x(t) := v_{\rightarrow} + \frac{1}{\#\mathcal{N}_i(t)} \sum_{j \in \mathcal{N}_i(t)} F[x_j(t) - x_i(t)], \quad (7)$$

$$v_i^y(t) := G[d_{i,\delta}^+(t)] - G[d_{i,\delta}^-(t)] + P \tan \alpha_i(t). \quad (8)$$

It is easy to see that only data currently accessible by USV i are used here.

Informally, Formula (7) is responsible for gathering the USVs on a common cross-section of the corridor and for driving them with the targeted x -speed $v_{\rightarrow}$. The first two addends in the r.h.s. of (8) are in charge for evenly distributing the USVs across the corridor. The duty of the last addend is to take into account possible inclination of the obstacles (including the corridor walls, see Section 7.7) with respect to the horizontal direction when distributing USVs across the width of the passageway.

The offered control algorithm is discontinuous; the closed-loop solution is meant in the Filippov's sense [58]. Since the controls are bounded, the solution is defined for all $t \geq 0$ [58]. Its uniqueness is a more intricate issue. We do not come into its discussion and address all solutions.

7. Main Results

7.1. Preliminaries

To state and discuss the main results, we need to introduce some symbols and terms. The following notations will be used:

- $h(E)$, length of the x -scope of the compact set E , i.e., $h(E) := \max_{r \in E} x(r) - \min_{r \in E} x(r)$;
- $\varphi(S) \in (-\pi/2, \pi/2)$, polar angle of a non-vertical segment S oriented in the ascending direction of x ;
- $\{S_q^{\pm}\}_{q \in \mathbb{Q}^{\pm}}$, collection of all globally upper/lower sides of all O_k 's (see Definition 3 and Figure 3).

Definition 4. A side S of an obstacle O_k is said to be friendly if the infinitesimal shift in the x direction brings any interior point of S to the free space. A friendly pair is a couple of adjacent friendly sides S_1, S_2 that are both globally either upper or lower for the obstacle O_k .

The side from Figure 6c is friendly, whereas those from Figure 6a,b are not friendly. In Figure 7, only the sides from the first and third pictures are friendly.

Finally, we put:

$$h_{\min} := \min \left\{ \min_{q \in Q^+} h(S_q^+); \min_{q \in Q^-} h(S_q^-) \right\}, \quad (9)$$

$$\hat{\alpha} := \max \left\{ \max_{q \in Q^+} \varphi(S_q^+); \max_{q \in Q^-} -\varphi(S_q^-) \right\}, \quad (10)$$

$$\beta_{\min} := \min_{\text{all friendly pairs}} \min \{ |\varphi(S_1)|; |\varphi(S_2)| \}, \quad (11)$$

$$g_{\min} := \min_{k \neq k'} \text{dist}[O_k; O_{k'}]. \quad (12)$$

Here, $\beta_{\min} := \infty$ if there are no friendly pairs, and $\hat{\alpha} \in [0, \bar{\alpha}]$, where $\bar{\alpha}$ is taken from (2). Since all sides $S_q^\pm$ are not vertical by Remark 1, we infer that $h_{\min} > 0$, $\hat{\alpha} \in (0, \pi/2)$, and $\beta_{\min} > 0$.

7.2. Key Theoretical Result

It refers to the constant $\bar{v}$ from (1), $\bar{\alpha}$, $v_{\rightarrow}$, R_{vis} from (2), α_* from (4), $\bar{F}$ from (5), $R_G, \bar{G}$ from (6), P from (8), $\varkappa$ from Figure 4b, w from Figure 1, the number of the USVs N , and the parameters δ and γ_x, γ_y that define the quantity (3) and the graph $\Gamma_i(t)$, respectively.

Theorem 1. Let Assumptions 1 and 2 and Equations (2), (5), and (6) hold, and the parameters of the control law be chosen so that the following inequalities are true:

$$\bar{F} < v_{\rightarrow}, \quad \frac{w}{N+1} < R_G < R_{\text{vis}}, \quad \gamma_x \tan \bar{\alpha} \leq \gamma_y, \quad (13)$$

$$(v_{\rightarrow} + \bar{F})^2 + [\bar{G} + P \tan \alpha_*]^2 \leq \bar{v}^2, \quad (14)$$

$$(v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} + \bar{G} < P \tan \alpha_*, \quad (15)$$

$$\sqrt{\min\{N\gamma_y; R_{\text{vis}}\}^2 + \varkappa^2} < g_{\min}/2. \quad (16)$$

Suppose that, at the start of the mission, the USVs are placed first outside the caves of the obstacles and lie in a vertical strip whose width γ is such that

$$\gamma < h_{\min}, \varkappa, \gamma_x, \delta, \sqrt{R_{\text{vis}}^2 - R_G^2}, \quad (17)$$

$$G[\gamma \tan \bar{\alpha}] < [v_{\rightarrow} - \bar{F}] \tan \beta_{\min}, \quad (18)$$

$$\bar{G} + G[\gamma \tan \hat{\alpha}] < P \tan \alpha_*, \quad (19)$$

$$\sqrt{\min\{N\gamma_y; R_{\text{vis}}\}^2 + (\varkappa + \gamma)^2} < g_{\min}/2. \quad (20)$$

Suppose also that initially different USVs are separated by no visored obstacle and have distinct ordinates. Then, the following statements are true:

- (i) The control $\mathbf{v}_i(t)$ generated by the proposed navigation law always satisfies the inequality from (1), as is required;
- (ii) The USVs sweep the scene with a speed of $v_{\rightarrow}$, as defined in Definition 1;
- (iii) Their x scatter $\varsigma(t) := \max_{i,j} |x_i(t) - x_j(t)|$ does not grow as time progresses;
- (iv) Any USV moves along the corridor in the targeted direction: $\dot{x}_i(t) > 0$ for almost all t ;
- (v) The USVs travel at distinct distances from the corridor walls: $y_i(t) \neq y_j(t) \forall i \neq j, t \geq 0$;
- (vi) The ordering of the USVs in the vertical direction does not change with time.

The proofs of this theorem and the subsequent Remarks 2–4 are given in Appendix A.

7.3. Discussion of the Key Theoretical Result

Due to (16), the visor's length $\varkappa < \frac{g_{\min}}{2}$. So, the visored obstacles are disjoint by (12).

In light of the mission objective, the initial placement of the USVs into a vertical strip is a reasonable decision. In practice, its realization often looks like putting the USVs into nearly the same place; then, in addition, no two USVs are separated by an obstacle, just like it is required in Theorem 1.

All assumptions on the initial deployment are met if, when moving along a free part of the corridor in a nearly perfect order, the team meets an obstacle course, and at this very moment, we set the clock to zero and start our consideration. In this case, the objective is to go through the obstacle course, while maintaining the line-up structure, and to restore even spacing and the required speed along the corridor after the course is left behind. In this regard, (iii) in Theorem 1 guarantees that the previously achieved “accuracy” ς of “lining up” never degrades, even when going through the obstacle course. By (5) and (7), the speed v_x deviates from the desired value $v_{\rightarrow}$ by at most $\bar{F}$, an observation which can be used to bound the speed error via choosing $F(\cdot)$. It is worth noting that as $\bar{F}$ decreases, neither of the affected requirements (13), (14), (15), and (18) from Theorem 1 is violated.

The statement (vi) means that, in the line formation, both the highest and lowest USVs retain their respective statuses. This may be of interest if there is a reason to use USVs with special roles or capabilities (e.g., those of the wall exploration or side-protection of the team) at these positions. Furthermore, no additions to the proposed control law are needed in scenarios where it is judicious to keep some special USVs hidden in the middle of the team, e.g., for protection purposes.

Theorem 1 sets forth achievable requirements. Indeed, the demands to R_G from (13) can be satisfied due to the second inequality in (2). Inequality (16) can be met by picking small enough $\varkappa > 0$ and γ_y , after which the last inequality in (13) can be ensured by choosing $\gamma_x > 0$ sufficiently small. By letting $\bar{F} \rightarrow 0$, $\bar{G} \rightarrow 0$, we make the first inequality in (13) true and reduce (14) and (15) to $\bar{v}^2 > v_{\rightarrow}^2 + P^2 \tan^2 \alpha_*$ and $v_{\rightarrow} \tan \hat{\alpha} > P \tan \alpha_*$, respectively. The system of the last two inequalities has a solution P, α_* if and only if

$$\bar{v}^2 > v_{\rightarrow}^2 + v_{\rightarrow}^2 \tan^2 \hat{\alpha} = \frac{v_{\rightarrow}^2}{\cos^2 \hat{\alpha}}, \quad (21)$$

which is true by the first formula from (2) since $\hat{\alpha} < \bar{\alpha}$. Hence, to meet (13)–(15), one can start with picking P and α_* so that $P \tan \alpha_* \approx v_{\rightarrow} \tan \hat{\alpha}$, $P \tan \alpha_* < v_{\rightarrow} \tan \hat{\alpha}$, and then proceed by choosing $\bar{F}, \bar{G} > 0$ small enough. At the end, it remains to take the functions $F(\cdot)$ and $G(\cdot)$ that satisfy (5) and (6) with the just chosen parameters $R_G, \bar{F}$, and $\bar{G}$.

We also note that (17)–(20) describe a non-empty set of γ 's by (6), (13), (15), and (16). For example, this set contains all small enough γ 's, but not necessarily only them.

Theorem 1 remains true if the corridor width w is replaced by its upper bound in (2) and (13), the angle $\hat{\alpha}$ by its known upper bound in (15) and (19), the angle $\beta_{\min}$ by its known lower bound in (18), and the distances $h_{\min}$ and $g_{\min}$ by their lower bounds in (17) and (16), (20), respectively.

Theorem 1 does not limit the choice of the parameter $\delta > 0$. If $\delta > R_{\text{vis}}$, then δ does not affect the control and so can be neglected everywhere since then $\mathcal{N}_{i,\delta}(t) = \mathcal{N}_i(t)$. The idea behind the use of δ stems from the fact that for $\delta < R_{\text{vis}}$, the set $\mathcal{N}_{i,\delta}(t)$ can be essentially smaller than $\mathcal{N}_i(t)$, which entails a reduction in the computational burden.

If, in violation of the assumptions of Theorem 1, $y_i(0) = y_j(0)$ for some $i \neq j$, then the objective is not achieved since $y_i(t) = y_j(t) \forall t \geq 0$. (The team acts as one of a reduced size and this smaller team gains its aim.) This can be fixed by slightly altering the control law. However, we drop discussion the respective details since the easiest tactics is to ensure that $y_i(0) \neq y_j(0) \forall i \neq j$ via proper placement of USVs at the start of the mission. Moreover, that assumption is met with a probability of 1 if the USVs are dropped in the scene according to a continuous probability distribution.

In terms of $z_i(t) := x_i(t) - v_{\rightarrow} \cdot t$, the control law (7) takes the form of the nonlinear *nearest neighbors rule* [59], which is used to achieve a consensus on the value of z_i among multiple agents:

$$\dot{z}_i = \frac{1}{\#\mathcal{N}_i(t)} \sum_{j \in \mathcal{N}_i(t)} F[z_j(t) - z_i(t)]. \quad (22)$$

Its convergence needs certain connectivity of the information-flow graph $\mathcal{G}(t)$ (we recall that its nodes $1, \dots, N$ are associated with the agents, and nodes i and j are linked if and only if $j \in \mathcal{N}_i(t)$). Most of consensus criteria merely assume that this graph enjoys a sort of connectivity for some reason, while leaving the reason an open issue. There are also studies on algorithms aimed at preserving the assumed initial connectivity; see, e.g., [15–20,60,61]. They are typically computationally expensive and rely on a fairly comprehensive knowledge of the scene.

On the contrary, the algorithm from this paper is computationally cheap and uses only local sensory information about the scene, whereas its convergence does not rely on the postulate of persistent connectivity or the assumption of the initial connectivity. The reason for its convergence is the proven fact that sooner or later the algorithm automatically gives birth to the connectivity.

7.4. Secondary Add-Ons to Theorem 1

Its assumptions are adopted throughout this section.

Remark 2. After some time has elapsed, the discontinuities of the control law (7) and (8) are being bypassed.

This excludes sliding-mode phenomena and their associates.

Remark 3. Whenever a USV is not at the front end of a visor V , the opinion of this USV on being above or below V corresponds to reality (and so the USV does not in fact displace V in its “mind”).

Now, we discuss the extension of the above findings to three special situations.

7.5. Dropout of Team Members

Suppose that, at a time $t = t_*$, several USVs abandon the mission and the others start ignoring them. By applying Theorem 1 to $t \geq t_*$, with a particular attention to (iii) and (v) at $t = t_*$, we see that the conclusion of Theorem 1 remains true for the remaining team if its size $N_* > wR_G^{-1} - 1$ (this keeps the second inequality in (13) true). With regard to the possibility of $R_G \approx R_{\text{vis}}$, the claim to N_* shapes into $R_{\text{vis}} > w/(N_* + 1)$ and means that the loss of several teammates does not break the sensorial connectivity of the team under its even distribution across the corridor.

Similarly, the team adapts to several successive dropouts if the final team size $> wR_G^{-1} - 1$.

7.6. Addition to the Team

We limit ourselves to a simple, though practical, scenario, where extra teammates (*newcomers*) are disembarked near some “old” ones (*oldies*). Let the newcomers appear at some time t_* not in the caves of the obstacles and within a vertical strip that contains the oldies and has a small enough width γ_{add} : (17)–(20) hold with $\gamma := \gamma_{\text{add}}$ (and the updated team’s size in (20)). Theorem 1 implies that, for $t \geq t_*$, its statements (i)–(vi) (and Remark 3) are valid if at $t = t_*$ the ordinates of the newcomers are pairwise different and differ from those of the oldies, and the obstacles do not obstruct the lines of sight among the USVs

(in fact, the last requirement can be relaxed; we omit the respective details since they are technically involved).

7.7. Non-Straight Corridors

The above findings nearly literally extend on more general corridors. The last sentence of the paragraph following Definition 3 hints at this possibility: since the upper and lower exteriors of the corridor are handled as obstacles, it suffices that these exteriors meet our requirements to the obstacles and the structure of the corridor makes the concepts of the corridor direction and its cross-section meaningful. These ideas are detailed in the following definition. To state it, we recall that a planar set is said to be *polygonal* if its boundary is contained in finitely many straight lines.

Definition 5. A directed simplistic corridor is an area $\mathfrak{C}$ such that there exists two polygonal sets $\mathcal{E}^+, \mathcal{E}^-$ and a directed straight line L with the following properties:

- (i) The sets $\mathfrak{C}, \mathcal{E}^+, \mathcal{E}^-$ are simply connected and form a partition of $\mathbb{R}^2$;
- (ii) There are x_+ and $w > 0$ such that, in a right-handed Cartesian frame whose abscissa axis is L , the parts of the sets $\mathfrak{C}, \mathcal{E}^+$, and $\mathcal{E}^-$ lying to the right of the abscissa x_+ are described by the inequalities $-w/2 \leq y \leq w/2, y > w/2$, and $y < -w/2$, respectively, along with $x \geq x_+$;
- (iii) Assumption 2 is true for $\mathcal{E}^+$ and $\mathcal{E}^-$.

This definition is illustrated in Figure 9.

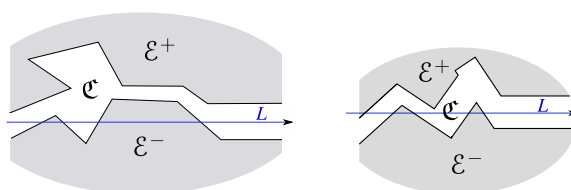


Figure 9. Non-straight simplistic corridors.

Remark 4. Let the corridor $\mathfrak{C}$ be simplistic. Suppose also that, in Section 7.2, all statements and formulas (including (9)–(12) and the definition of $\bar{\alpha}$ from (2)) that refer to “obstacles” take into account $\mathcal{E}^+$ and $\mathcal{E}^-$. Then, Theorem 1 remains true.

The assumption of Theorem 1 on starting outside the caves can also be relaxed. We illustrate this in Section 8 by computer simulations, with omitting theoretical discussion since it is technically cumbersome.

7.8. Some Hints to Relax Assumption 2

Assumption 2 excludes the situations like those in Figure 2a,c. However, they sometimes can be reduced to a case where Assumption 2 is met. To this end, the USVs may imaginarily endow the obstacles with extra fictitious parts. The price for this is that in the opinion of the USVs, the free space shrinks.

Now, we outline the basic idea for two hints of this kind, leaving the elaboration of their details as a topic of further research. Every hint reduces the number of critical rays by one. If the obstacle has many such rays, their successive processing is recommended.

Adding a fictitious cusp to a vertical front side S of an obstacle is illustrated in Figure 10, where $\Delta > 0$ is a free parameter and the cusp is depicted in green.

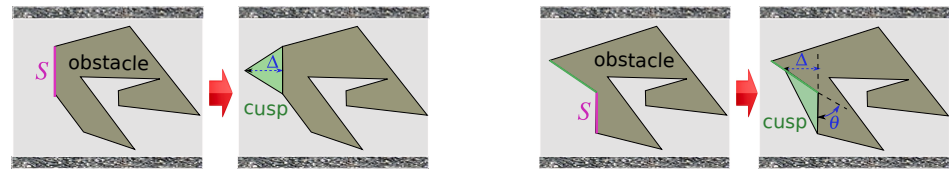


Figure 10. Adding a fictitious cusp in front.

The case where the both sides adjacent to S go up the corridor is of no interest: this means that the USV is in a cave, which should have been prevented by properties of the scene and algorithmic hints. To build the fictitious cusp in a timely manner, the USV must have a full view of S whenever the cusp may affect the control. So, the length s of S and Δ should not be excessively large. Also, it is needed that the cusp points should be visible whenever they fall in the sensing range R_{vis} , as it is given to the controller. Hence, the USV should “see” not only these points but also the entirety of S . To create such situation, the controller may be affectedly given a value of R_{vis} that is lesser than the real one.

Filling up unacceptable caves is illustrated in Figure 11. In Figure 11b–d, $\Delta > 0$ is a free parameter. Figure 11d addresses the case where the obstacle’s side that ought to support the cusp is too short to accommodate its entirety and so the cusp is cut at the respective end of this side. The horizontal mirror images of all pictures from Figure 11 should be also included in the list of illustrations. Like in the case of vertical front sides, the USV must have the full view of the gate to the cave whenever the cave affects the control. Meanwhile, more may be needed. For example, if different obstacles are not somehow discernable (e.g., do not vary in color), then the entire cave is welcome to be visible from its entrance to differentiate it from a passage between obstacles, unless counting on luck. In Figure 11, only the cave from Figure 11d has this property.

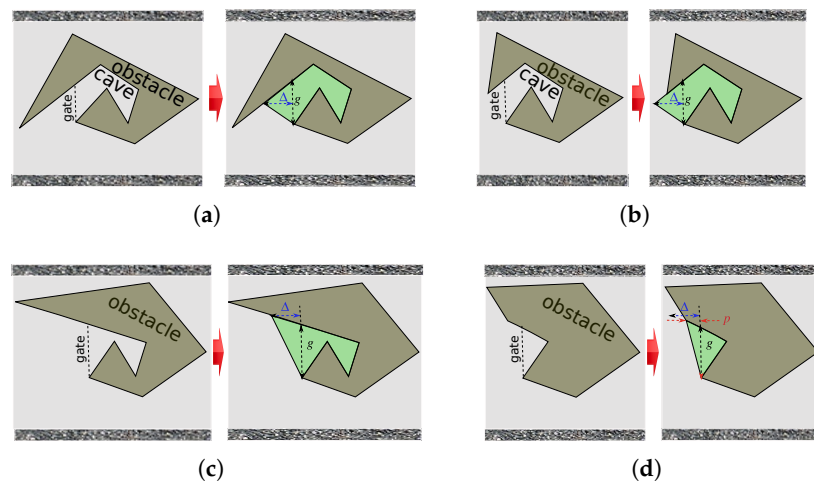


Figure 11. Filling up an unacceptable cave. (a, b, c) Various examples of obstacles with caves that can fully accommodate the cusp; (d) The case where the obstacle’s side that ought to support the cusp is too short to accommodate its entirety and so the cusp is cut

8. Computer Simulation Tests

8.1. Numerical Experiments

The experiments described in this subsection were implemented in a Jupyter Notebook using the Python programming language. For visualization and vector operations, the Pygame and NumPy libraries were used, respectively. The walls of the corridor were defined as graphs of functions of the abscissa x from Figure 1. The obstacles were described by two functions representing the “upper” and “lower” parts of their boundaries, respectively.

The obstacles whose boundaries cannot be separated into such two parts (like the obstacles in Figure 17) were decomposed into sub-obstacles that allow this separation (in Figure 17, this decomposition is outlined by the little “seams” on the obstacles). The discretization of the continuous-time kinematic model of the robots (1) was performed using the Euler method; the discretization step equals the control update period τ shown in Table 1.

Apart from the validation of the proposed control laws (7) and (8), the goals of the simulation studies described in this subsection include the assessment of its applicability in circumstances outside the bounds imposed in the theoretical part of the paper. In particular, the tests involve erroneous position measurements, controller parameters that do not quite satisfy the assumptions of Theorem 1, scenes with non-polygonal and moving obstacles, obstacles with vertical front sides, and situations where robots start in the caves of the obstacles. Meanwhile, the employed kinematic model is close to the theoretical model (1). Testing the algorithm against the challenges from more complex dynamics and other imperfections of the real world is discussed in Section 8.2 and 9. Various validation means (numerical simulation, Gazebo simulation, real-life experiments) are applied independently, with the main objective to cover a wider range of scenarios.

The numerical values of the key parameters used in the tests from Section 8.1 are shown in Table 1. We employed a linear function $G(\cdot)$ with saturation at the threshold R_G , Table 1 displays only its linear part, as well as the control update period τ . The parameters from Table 1 are not fully compliant with the assumptions of Theorem 1; this is done to show that the domain of practical convergence of the offered control law is wider than its region of convergence unveiled theoretically.

Table 1. Numerical values used in the computer simulation tests.

$\bar{v} = 6.0 \text{ m/s}$	$R_{\text{vis}} = 1.5 \text{ m}$	$v_{\rightarrow} = 1.0 \text{ m/s}$
$P = 2.0 \text{ m/s}$	$\delta = R_{\text{vis}}/\sqrt{2}$	$\alpha_{\star} = \pi/4$
$F(x) = \frac{7.8x}{\sqrt{1+x^2}}$	$R_G = R_{\text{vis}}$	$G(d) = 0.78 \cdot d \text{ 1/s}$
$\gamma_x = R_{\text{vis}}/4$	$\gamma_y = R_{\text{vis}}/2$	$\tau = 20 \text{ ms}$

Position measurements were plagued with unbiased additive noises uniformly distributed within a radius of 0.01 m. This is a fairly big inaccuracy relative to the typical inter-USV distances of $\approx 0.3\text{--}1.0 \text{ m}$ observed in our experiments: e.g., many present-day sensors ensure a precision of several mm in that range of measurements. The multimedia of all experiments are available at <https://drive.google.com/drive/folders/1ngTYqiY1V-2-sEI6EjcMrFmMAEFSMWuM?usp=sharing> (accessed on 18 February 2025).

In Figures 12–19, the positions of the USVs and the orientations of their velocities are demonstrated with the triangles; the time step between the snapshots equals 0.4 s; the trajectories of the USVs are shown as blue lines, while the boundaries of the corridor and obstacles are shown as green lines. Except for the first snapshot, the robotic team clusters near a cross-section of the corridor, which makes the snapshots easily recognizable. These snapshots allow us to evaluate the speeds of the USVs.

In Figure 12, eleven USVs are tasked with performing a dense sweep of a corridor with slightly broken walls and three obstacles. Beginning from an initial patch of the corridor with a diameter of $\approx 1.5 \text{ m}$, the team accelerates to the desired speed, aligns across the corridor, and achieves an even self-distribution within about 4.0 s. This distribution remains stable until around 7.0 s, at which time the USVs encounter the first obstacle. While navigating through the obstacle course, the USVs maintain their “line-up” formation and targeted horizontal speed, in spite of the fact that USVs separated by obstacles lose information about one another. Meanwhile, the team dynamically breaks into subgroups,

each utilizing different pathways through the obstacles. The number and composition of the subgroups change over time, following the sequence: $\{11\} \rightarrow \{2, 9\} \rightarrow \{2, 7, 2\} \rightarrow \{2, 3, 4, 2\} \rightarrow \{5, 4, 2\} \rightarrow \{5, 6\} \rightarrow \{11\}$. This progression involves both splitting and merging of subgroups. As shown in Figure 12, the USVs do not collide with the walls of the corridor, the obstacles, and one another. By approximately 30 s, the team re-establishes the “cross-section” formation and maintains the requested horizontal speed $v_{\rightarrow}$, despite the disruptions caused by obstacles. Thus, the posed objective is attained with high accuracy.

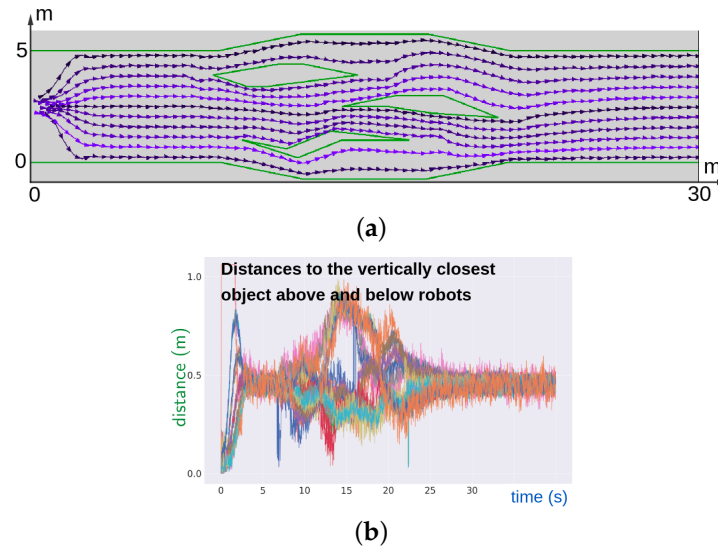


Figure 12. Eight USVs traverse an obstacle course. (a) USVs trajectories with snapshots in the corridor with obstacles; and (b) Graph of the distances to the vertically closest object above and below USVs.

The purpose of the experiment from Figure 13 is to test the hint of adding a fictitious cusp; this hint is illustrated in Figure 10a and aims to handle obstacles with a vertical front side. In the opinion of the controller, the sensing range of the robot was artificially reduced down to 65% of its true value. Starting their mission in a circle with a diameter of ≈ 1.5 m, the robots achieve the required speed and uniformly distribute themselves across the width of the corridor in ≈ 5.0 s, having traveled ≈ 5.0 meters along the corridor. This uniform distribution is maintained until ≈ 15.0 s, when the robots encounter an obstacle with a vertical front side. To avoid a collision, the concerned robots independently expand the obstacle by adding an imaginary cusp to this side, as shown in Figure 10a, and move along the boundary of this cusp. After bypassing the obstacle, it takes ≈ 10 seconds to restore the formation. The results of this experiment do not significantly differ from those described above, and the control objective is again achieved with a good precision.

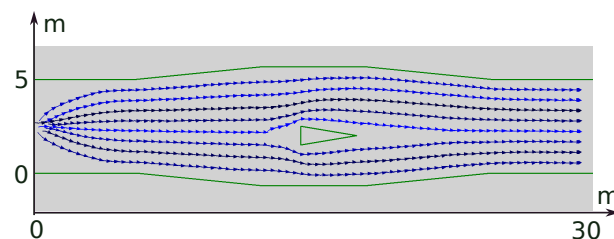


Figure 13. Eight robots, one obstacle with a vertical front side.

In the scenario from Figure 14, a group of eight robots has to pass through a corridor with oval obstacles. At the start of the mission, the robots do not have enough space and time to attain the required speed and even distribution over a cross-section of the corridor, so long as they almost immediately stumble upon the first band of three obstacles. It takes

≈ 10 s for them to overcome this band. Then, they once again do not have enough time and space to distribute themselves uniformly across the corridor, this is due to the second band of obstacles. Overcoming this band takes ≈ 10 s, after which the robots achieve uniform self-distribution across the corridor in ≈ 5 s, and then stably maintain the required formation, as confirmed by Figure 15. The interest in this experiment partly stems from the fact that the oval obstacles violate Assumption 1. However, the obtained results demonstrate that the algorithm is applicable to scenarios with non-polygonal obstacles and may attain the control objective with good exactness.

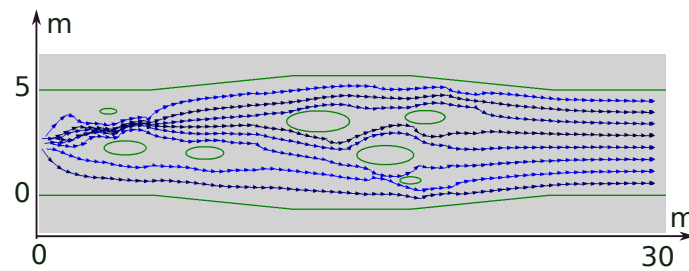


Figure 14. Eight robots, scenario with oval obstacles.

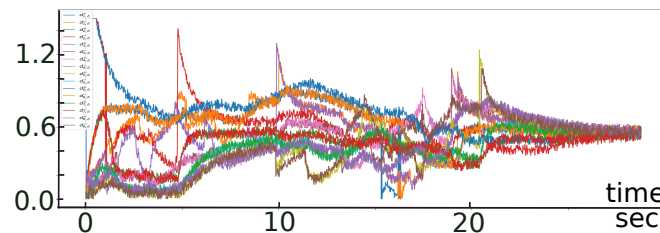


Figure 15. Graph of the lateral distance to the nearest upper/lower object $d_{i,\delta}^{\pm}(t)$.

Figure 16 illustrates an experiment in which 11 robots should sweep a corridor with an obstacle that oscillates across the corridor. At $t \approx 3.0$ s, the group lines up across the corridor and achieves a nearly even inter-robot spacing; in the meantime, the obstacle is displaced slightly downward. At $t \approx 7.8$ s, the group begins to notice the obstacle, whereas the latter reaches its lowest position and starts changing direction. By $t \approx 11.2$ s, the group splits into two subgroups: three robots “decide” to bypass the obstacle from below, and the rest of the team prefers to go above it; during the entire obstacle avoidance maneuver, the obstacle moves upward. As a result, the free space available to the upper group of eight robots becomes considerably constricted at $t \approx 13.6$ s. At $t \approx 19.8$ s, the group finishes bypassing the obstacle and merges into a single unit. Finally, at $t \approx 27.2$ s, the group restores an uniform distribution across the corridor, thereby achieving the control objective. Thus, we observe that the mission is successfully accomplished even in the presence of a moving obstacle.

The experiment addressed by Figure 17 illustrates the remark from the last paragraph in Section 7.7: the algorithm maintains efficiency in situations where USVs start inside caves. Initially, twelve USVs are distributed in three groups out of four. Two of them start in the caves of the obstacles O_1 and O_2 , respectively, and the third group is launched in between O_1 and O_2 . In ≈ 2 s, a nearly perfect alignment across the corridor is achieved within each group, along with an approximately even distribution of each group over the width of the currently used “left-to-right passage” between the obstacles’ components. Meanwhile, the sensorial contact between the groups is blocked by the obstacles. As a result, there is no progress in narrowing the x gaps among the groups (these gaps are highlighted with dashed pink rectangles). At $t \approx 7$ s, the “upper” and “middle” groups merge, and ≈ 2 s later, they appear to be close to a common moving cross-section of the corridor. Meanwhile, their alignment with the “lower” group $\mathcal{G}_\downarrow$ still remains at the nearly

initial level and poor. This does not come as a surprise since that group is still out of the sensorial contact. At $t \approx 11$ s, the united “upper” group splits into two subgroups $\mathcal{G}_1$ and $\mathcal{G}_2$, which “decide” to bypass the obstacle O_3 from above and below, respectively. Initially, $\mathcal{G}_2$ is out of contact with $\mathcal{G}_\downarrow$ (first because of the blocking effect of O_2 and then since the gap between $\mathcal{G}_2$ and $\mathcal{G}_\downarrow$ exceeds the sensing distance). During this period, $\mathcal{G}_1$ and $\mathcal{G}_2$ retain a nearly perfect mutual alignment across the corridor, though they are separated with O_3 . The control law stimulates $\mathcal{G}_2$ and $\mathcal{G}_\downarrow$ to reduce the gap between them. As a result, they come into contact and the resulting united group aligns across the corridor and achieves a nearly even self-distribution over the width of the employed passage at $t \approx 13.0$ s. A side effect of these is that $\mathcal{G}_2$ loses the alignment with $\mathcal{G}_1$. Only after the obstacle course is left behind (at $t \approx 19$ s) and the gap between $\mathcal{G}_1$ and the remainder of the team is reduced to below the sensing range (at $t \approx 21$ s), the alignment of the entire team and its even self-distribution across the corridor is attained at $t \approx 25$ s. Once more, we observe that the mission is carried out successfully and with a good accuracy.

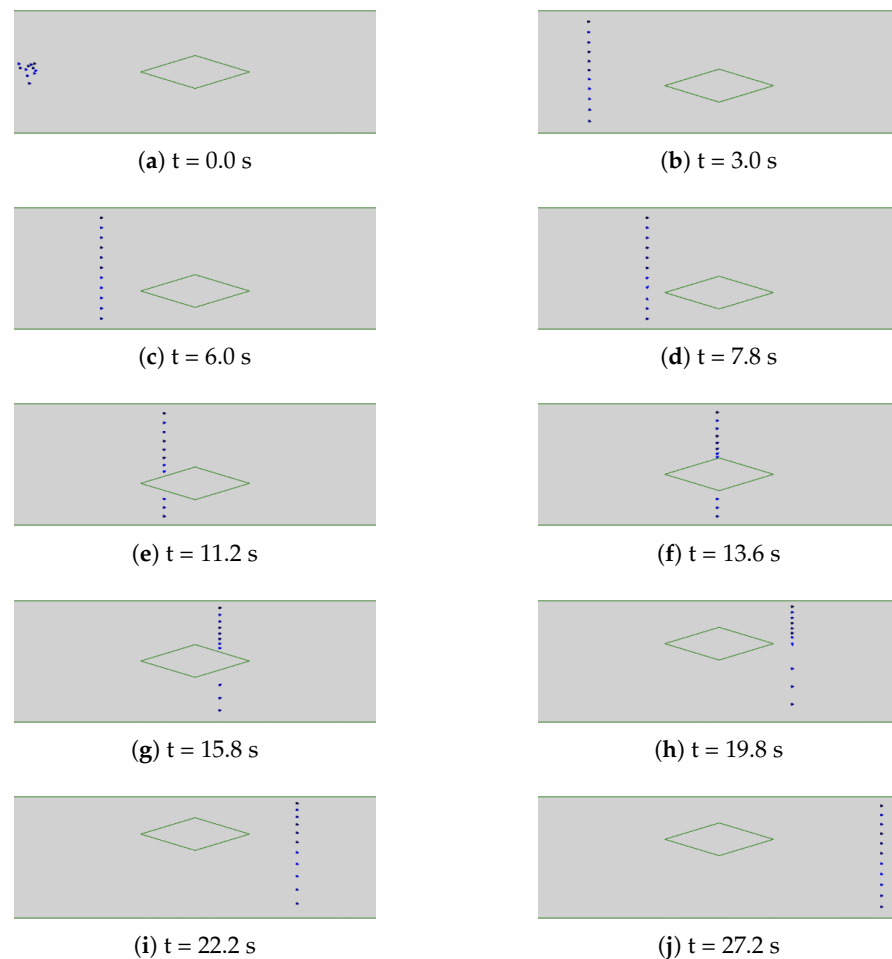


Figure 16. Scenario with a moving obstacle.

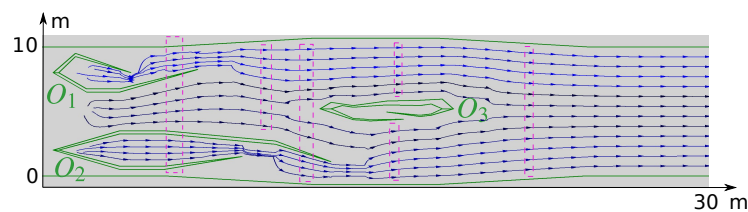


Figure 17. Starting in caves, twelve USVs.

The last two experiments test the capacity of the algorithm to cope with changes in the team membership. In the scenario from Figure 18, the team of 11 USVs loses 5 USVs in the middle of the corridor length. After this happens, the remaining six USVs run the control rule “as usual”, ignoring the missed peers. By Figure 18, the result successfully traverses the obstacle course and achieves optimal (for the downsized team) self-spreading across the corridor. In Figure 19, six USVs encounter five motionless red “newcomers”. As soon as a newcomer detects that a USV overtakes it (in projection onto the x axis), it starts running the control rule and so begins to move. At the “time of meeting”, obstacles block the line of sight between some USVs. As can be seen, this does not bring any trouble: the united team reasonably uses the available passages between the obstacles, eventually leaves the obstacle course behind, and achieves the optimal deployment across the corridor. Thus, the algorithm demonstrates robustness to changes in the team membership.

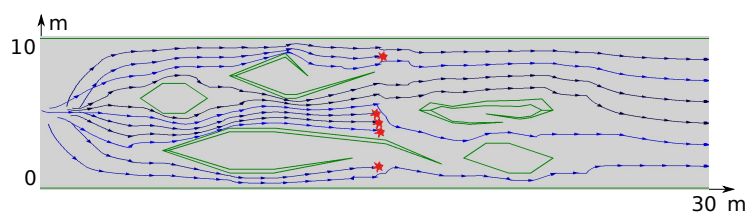


Figure 18. A team of eleven USVs unexpectedly loses five ones.

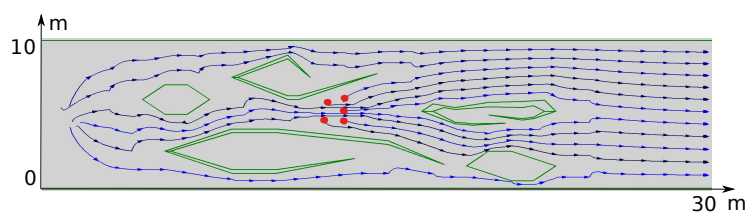


Figure 19. Five USVs join a team of six ones.

8.2. Gazebo Experiments

To test the control law against the challenges from the more complex dynamics of the robots, some experiments were carried out using the realistic 3D robotics simulator Gazebo 11 (<https://gazebo.org/home>, accessed on 18 February 2025) and ROS 2 (<https://www.ros.org/>, accessed on 18 February 2025). This simulator provides high-quality physics engines for modeling the robot and environment and emulates realistic sensor readings.

Two-wheeled differential drive robots are used as the robotic platform. To model it, the DiffBot package <https://github.com/ros-mobile-robots/diffbot> (accessed on 18 February 2025) is employed; it has been slightly modified subject to the considered application scenarios. A two-level control system was designed to control each robot. The control law (7) and (8) acts at the higher level to generate the desired velocity vector. This vector is then fed into the lower-level control module, whose role is to generate control signals for the left and right wheels in order to achieve the desired velocity. This module is based on PD controllers and is developed using standard control system design methods.

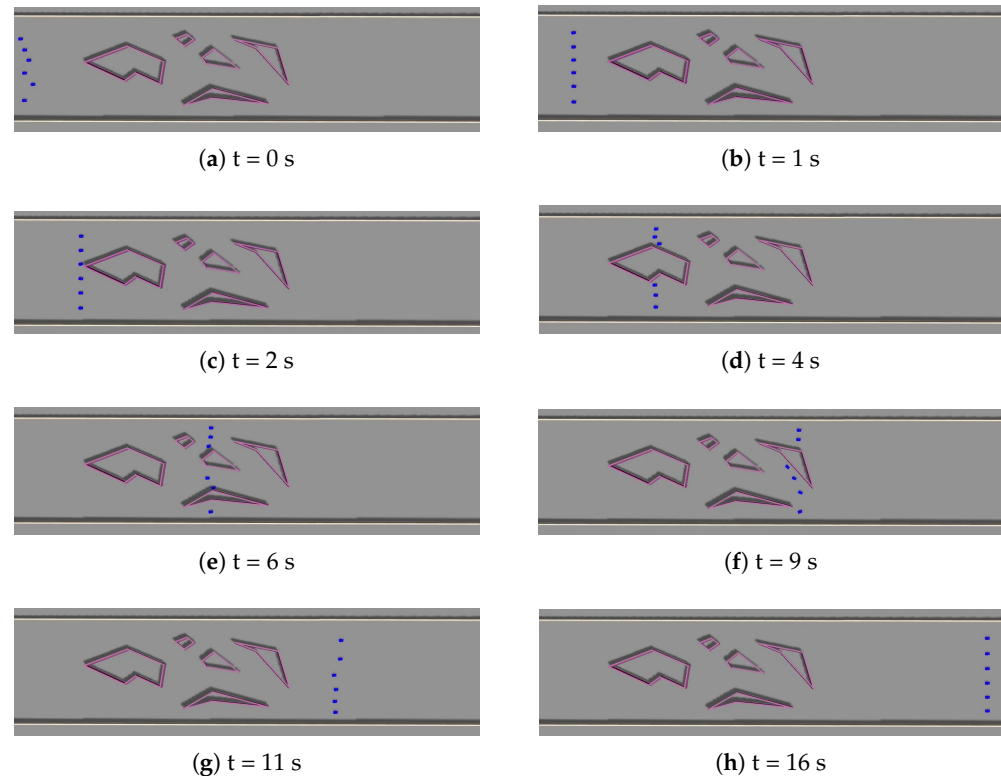
Partly in order to better exhibit the region of the control law convergence and to examine its robustness against controller tuning errors, some parameters of the algorithm and scenario are altered in Section 8.2 as compared with Section 8.1.

The parameter values used in Gazebo simulation are reported in Table 2.

Table 2. Numerical values used for simulations with Gazebo.

$\bar{v} = 1.5 \text{ m/s}$	$R_{\text{vis}} = 4.0 \text{ m}$	$v_{\rightarrow} = 0.3 \text{ m/s}$
$P = 2.0 \text{ m/s}$	$\delta = 3.4m$	$\alpha_{\star} = \pi/4 \text{ rad}$
$F(x) = \frac{0.5x}{\sqrt{1+x^2}}$	$R_G = R_{\text{vis}}$	$G(d) = 0.1 \cdot d \text{ 1/s}$
$\gamma_x = R_{\text{vis}}/4$	$\gamma_y = R_{\text{vis}}/2$	$\tau = 200 \text{ ms}$

The results of a typical experiment are shown in Figure 20.

**Figure 20.** Six differential drive robots passing through a band of five obstacles.

The robots achieve the required speed and uniform distribution across the corridor in just ≈ 1.0 second; see Figure 20b. The group of robots encounters the band of obstacles at ≈ 3.0 s, as seen in Figure 20b. While overcoming the band of obstacles, the number of subgroups changes over time as follows: $(6) \mapsto (3-3) \mapsto (6) \mapsto (3-2-1) \mapsto (5-1) \mapsto (3-2-1) \mapsto (2-4) \mapsto (6)$. Meanwhile, Figure 20c–f shows that, during the traversal of the obstacle band, the robots never collided with an obstacle; they also did not collide with each other or with the corridor walls throughout the experiment. Although the robots are noticeably disorganized upon exiting the obstacle band in Figure 20g, they quickly restore perfect order in Figure 20h. Thus, the experiment demonstrated that, for robots with more complex kinematics, the proposed control law is suitable as a high-level control module within a hierarchical two-level control system.

A video recording of the experiment is available at the following link: <https://drive.google.com/drive/folders/1ngTYqiY1V-2-sEI6EjcMrFmMAEFsMWuM?usp=sharing> (accessed on 18 February 2025).

The one in Figure 21a is mounted on the Rover 5 USV platform (<https://www.sparkfun.com/products/retired/10336>, accessed on 18 February 2025); the USV in Figure 21b is similar in architecture to the DiffBot (<https://github.com/ros-mobile-robots/diffbot>, accessed on 18 February 2025). Simple gearbox and motors are mounted on the

wheeled chassis. Under a low stall torque situation, the acceleration of the chassis as a whole proved to be inappropriately small, presumably due to features of the employed simple gearbox. This entails a visible problems with the USV performance. To compensate for this effect, a special control loop is included in the above situation, which artificially increases the average current in the motor coil.

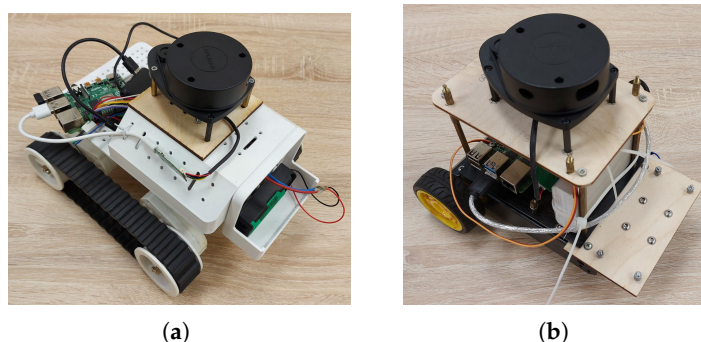


Figure 21. Laboratorial USV with (a) crawler and (b) wheeled chassis.

9. Results of Real-World Tests

These were performed using the two USVs shown in Figure 21.

Every USV is equipped with two (i.e., left and right) DC brush motors, and only the motors differ between the USVs. One of them employs the motor driver L298N, the other makes use of the Motor Shield (<https://amperka.com/>). The algorithm (7) and (8) is implemented onboard on the basis of a Raspberry Pi 4B+ computer running under Linux. The motors are driven with voltage pulse signals; their pulse amplitude is maximal. Meanwhile, the speed profile generated in accordance with (7) and (8) is converted into the pulse ratio for every motor by means of a simple low-level regulator. Thus, the two USVs are subjected to a common higher-level control rule, whose hardware implementation is different for them. As a result, the experiments somewhat examine the robustness of the rule (7) and (8) against such a difference. The basic parameters of the control law are taken from Table 1.

Each USV uses a range measuring scanning laser Slamtec RPLidar A1. It is mounted on the top of the USV in order to avoid blind spots. The Lidar readings are updated with a frequency of 10 Hz, and the motor control loop operates at a frequency of 50 Hz (this imbalance aids to smoothen the USVs' path). The computer, motors, and Lidar are separately powered by their own batteries. This separation is partly motivated by the wish to avoid knocking down the work of the Lidar because of disturbances from the motor sharing the same power source. The control software was implemented in the Thonny programming environment using the Python programming language. The `pyrplidar` library was used to obtain data from the LiDAR, and the `RPi.GPIO` library was used for motor control. At the level of the software implementation of the basic control law, extra measures were taken to encourage the robots to align with the direction of the corridor.

The multimedia of four real-life experiments are available at https://drive.google.com/drive/folders/1pRoDqK7XrvcZExlk-PTmdw_nc3zVeTBt?usp=share_link (accessed on 18 February 2025).

In Figure 22, the USVs go through a corridor $\mathcal{C}$ with two obstacles made of corrugated white paper. Their initial deployment differs from the desired one. At $t \approx 4.0$ s, they achieve a nearly even distribution over the width of the corridor but have not had enough time to equalize their coordinates along $\mathcal{C}$. A little later, the wheeled USV starts bypassing obstacle 1 from the right, so that, at $t \approx 10.0$ s, the USVs find themselves within a common

passageway between obstacle 1 and the right side of $\mathcal{C}$. At $t \approx 15.0$ s, this obstacle is left behind, the USVs nearly line up across $\mathcal{C}$, and are in the process of increasing the distances between themselves and to the right side of $\mathcal{C}$, which is needed to achieve an even distribution across $\mathcal{C}$. This process is interrupted when the line of sight between the USVs becomes blocked by obstacle 2. Then, the USVs do not “see” each other, and the control law has to drive each of them to the center of the employed passageway. This goal is achieved with good precision at $t \approx 18.0$ s. After obstacle 2 is left behind, the progression towards the even distribution resumes. As a result, at $t \approx 32.0$ s, the targeted formation is built with good accuracy; from this time forth, it is stably maintained.

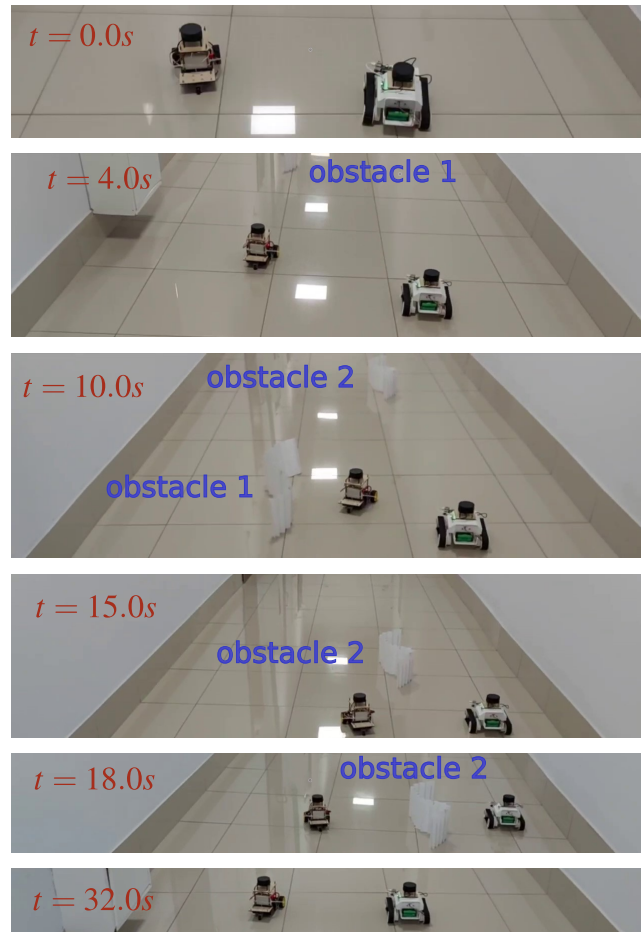


Figure 22. Two USVs go through a corridor with two obstacles.

These facts are confirmed by Figure 23, which exhibits (slightly post-processed) telemetry data from the two Lidars. In Figure 23, the blue and orange graphs display the distance from the USV to the nearest object (either a corridor wall, or an obstacle, or another USV) to the left and right, respectively, that is found in the narrow beam issued from the USV perpendicularly to the corridor direction. Though these data are somewhat damaged by experimental artifacts and brief disruptions in the regular operation of the distance evaluation system (which are manifested as short large spikes in the graphs), it confirms that, first, the USVs do not collide with each other, the corridor walls, and the obstacles and second, evenly distribute themselves across the width of the corridor (with the maximal positional error $\lesssim 4.0$ cm) after the obstacle course is left behind. Thus, the control law copes with the mission and demonstrates a reasonable performance.

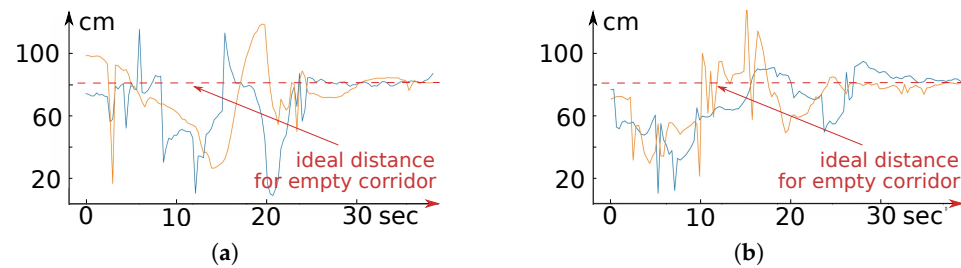


Figure 23. Telemetry data for (a) crawled and (b) wheeled USVs.

Overall, these and our other experiments (reported at the previously indicated site) demonstrated the robustness of the proposed control law against real-life non-idealities, including sensor noises, actuator errors, and mismatches between the theoretical model of the USV and its real-world kinematics and dynamics.

10. Conclusions and Future Work

A distributed and computationally cheap control rule was offered such that, when it is individually run by every USV from a team of an unknown size, the team autonomously organizes itself into the densest net-like straight barrier across an unknown corridor. This holds even if the corridor is partly cluttered with unknown polygonal obstacles. Moreover, the rule guarantees that, eventually, the barrier goes along the corridor with a pre-specified speed and in the given direction, and the USVs evenly distribute themselves across the width of the corridor in its obstacle-free parts. Furthermore, the straight cross-section formation of the robots is not broken in all parts of the corridor, including those cluttered with obstacles. The suggested control law uses only local sensory data within a finite distance. By directly converting the observations into control in a reflex-like fashion, this law demands minimal computational resources. Regarding the simplification of the control system architecture, it was demonstrated that, under certain circumstances, the task of obstacle avoidance can be entrusted to the control module responsible for distributing the robots across the width of the corridor. The proposed algorithm was shown to be robust against the dropouts of teammates and admissions of newcomers. It was also proven that its convergence does not rely on the postulate of persistent informational connectivity of the team or the assumption on the initial connectivity. Its convergence and performance were justified via mathematically rigorous non-local convergence results and were confirmed via intensive computer simulation tests and real-world experiments. This validation was performed and the applicability of the proposed approach was demonstrated for a whole variety of scenarios, including those outside the bounds of the theoretical analysis. In particular, the method works well in certain scenarios with moving or non-polygonal obstacles, and if USVs start in “caves” inside obstacles; also, it is somewhat robust against controller tuning errors.

The practical design of navigation laws often follows through a two-stage hierarchy, where control at the kinematic level generates a reference signal, which is then tracked by standard, more or less, controllers. The developments of this paper are attributed to the first stage and are, to a certain extent, about to set forth a sort of paradigm and provide a proof of concept. Implementation issues concerned with the second stage in a context somewhat similar in spirit to that from this paper are addressed in, e.g., [62,63].

Future work includes the elaboration of the details concerned with higher-order models of the USVs and relaxation of the assumptions about the obstacles. Our agenda also includes studies of scenarios where several barriers should be formed that consecutively follow one another with a given step. Extensions to 3D environments are also on the way.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/drones1010000/s1>, Video S1: animation of the experiment from Figure 17; Video S2: animation of the experiment from Figure 19; Video S3: animation of the experiment from Figure 18; Video S4: animation of the experiment from Figure 22.

Author Contributions: P.K.: Conceptualization, software, validation, formal analysis, investigation, writing—original draft preparation. A.M.: Conceptualization, methodology, data curation, formal analysis, writing—review and editing, supervision, project administration, funding acquisition. K.G.: software, validation, visualization. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

USV	Unmanned surface vehicle
a.a.	"for almost all"
Conventions and notations	
Positive angles	counted counterclockwise
$:=$	"is defined to be", "is used to denote"
"segment of a line"	its part between two points
"segment"	"segment of a straight line"
$[a, b]$	segment bridging the points a, b
$(a, b]$	$[a, b]$ minus a
$[a, b)$	$[a, b]$ minus b
(a, b)	$[a, b]$ minus a and b
$\#A$	number of elements in the set A
$\text{sgn}(a)$	signum function, $\text{sgn}(0) := 0$
∂O	boundary of the set O
$x(p), y(p)$	coordinates of point p
$\emptyset$	empty set
$\mathcal{C}$	handled corridor
w	width of the corridor
N	number of the robots
O_k	k th obstacle
r_i	location of robot i
x_i, y_i	coordinates of robot i
v_i	velocity vector of robot i
$\bar{v}$	upper bound on the robot's speeds
$v_{\rightarrow}$	desired speed
R_{vis}	sensing distance
$\mathcal{F}_k^{\pm}$	upper/lower fences of obstacle O_k
$\bar{\alpha}$	upper bound on the polar angles of the segments in upper and lower fences
$\varkappa, \delta, \gamma_x, \gamma_y, \alpha_*, P, R_G, \bar{G}, \bar{F}$	controller parameters
$F(\cdot), G(\cdot)$	functions used to build the controller

Appendix A. Proof of Theorem 1 and Remarks 2, 3, and 4

We study the motion of the team driven by the proposed control law. The assumptions of Theorem 1 are adopted throughout the appendix. Furthermore, the x distance is the distance between the abscissas of two points; the y distance is defined likewise.

Lemma A1. *The statement (i) of Theorem 1 is true.*

Proof. By (4), $|\alpha_i(t)| \leq \alpha_*$ and so $|\tan \alpha_i(t)| \leq \tan \alpha_*$. By invoking (5) and (7), we see that $|v_i^x(t)| \leq v_{\rightarrow} + \bar{F}$. Simultaneously, (6) and (8) imply that $|v_i^y(t)| \leq \bar{G} + P \tan \alpha_*$. The proof is completed by (14). $\square$

To state the next lemma, we put $v_x := v_{\rightarrow} - \bar{F}$ and note that $v_x > 0$ thanks to (13).

Lemma A2. *The statements (iii) and (iv) of Theorem 1 are true. Moreover, $\dot{x}_i(t) \geq v_x$ a.a. t .*

Proof. (iii) We start with invoking the notation $z_i(t) := x_i(t) - v_{\rightarrow} \cdot t$ from (22). By virtue of the Danskin's theorem [64], the function $M(t) := \max_i z_i(t)$ is absolutely continuous and $\dot{M}(t) = \dot{z}_{i_t^0}(t)$ a.a. t , where i_t^0 is a maximizer for $\max_i z_i(t)$. By (7) and the definition of the Filippov's solution [58], $\dot{z}_{i_t^0}(t)$ lies in the convex hull of the values that are the limits of

$$\frac{1}{|\mathcal{N}_{i_t^0}(t)|} \sum_{j \in \mathcal{N}_{i_t^0}(t)} F[z_j(t) - z_{i_t^0}(t)] \quad (\text{A1})$$

as the perturbed system's state $\{r_i(t)\}_{i=1}^N$ goes to the true one, respecting certain restrictions. However, they cannot influence the limit values of the addends, which are exactly as shown in (A1) and are nonpositive by (5). Hence, $\dot{M}(t) = \dot{z}_{i_t^0}(t) \leq 0$ a.a. t and $M(t)$ is a non-increasing function. Likewise, $m(t) := \min_i z_i(t)$ is a non-decreasing function. It remains to note that $\zeta(t) = \max_{i,j} |z_i(t) - z_j(t)| = M(t) - m(t)$.

(iv) By (5), the second addend in (7) does not exceed $\bar{F}$ in absolute value. It remains to invoke the first inequality from (13). $\square$

Corollary A1. $|x_i(t) - x_j(t)| < \gamma \forall t \geq 0, i, j$, where γ is taken from Theorem 1.

Lemma A3. *If USV i is on a visor V at time τ , then for $t \approx \tau, t > \tau$, it is outside V and is actually on the side of V that was downloaded at random into its “mind” at $t = \tau$.*

Proof. Let USV i “think” that it is below V at $t = \tau$; the opposite case is handled likewise. Then, $d_{i,\delta}^+(\tau+) = 0$ and $\alpha_i(t) := -\alpha_* \forall t < \tau, t \approx \tau$ due to (3) and (4). It remains to invoke (6) and (8) and observe that

$$\overline{\lim}_{t \rightarrow \tau-} v_i^y(t) = \overline{\lim}_{t \rightarrow \tau-} \left[G[d_{i,\delta}^+(t)] - \underbrace{G[d_{i,\delta}^-(t)]}_{\geq 0} + P \tan \alpha_i(t) \right] \leq G(0) - P \tan \alpha_* \stackrel{(6)}{=} -P \tan \alpha_* < 0.$$

Lemma A4. *Let $[0, \tau]$ be a time interval during which the USV i does not collide with the obstacles: $r_i(t) \notin O_k \forall k$. Then, the following statements hold:*

- (i) For $t \in [0, \tau]$, USV i is not in the caves of any obstacle O_k ;
- (ii) If there is a point of an obstacle O_k that lies above (below) $r_i(t)$, the closest $p_i^\pm(t)$ of such points belongs to the lower (upper) fence of O_k .

Proof. (i) Suppose that (i) is untrue. Since USV i is initially outside the caves by the assumptions of Theorem 1, there is time $\vartheta \in (0, \tau]$ and an obstacle O_k such that $r_i(\vartheta)$ is in a cave of O_k , whereas $r_i(t)$ is not in its caves for $t \in [0, \vartheta)$. By invoking the definition of the

cave, we see that the ray $\mathcal{R}_+[r_i(t)]$ does not intersect O_k for $t \in [0, \vartheta)$, $t \approx \vartheta$, or the same is true for the ray $\mathcal{R}_-[r_i(t)]$. We focus on the first situation, and the second one is treated likewise. We vertically extend $\mathcal{R}_+[r_i(t)]$ down from $r_i(t)$ until hitting O_k at some point $h(t)$ and thus obtain an O_k -free upward ray $\mathcal{R}'_+(t)$. Since $r_i(\vartheta)$ is in a cave, both rays $\mathcal{R}_\pm[r_i(\vartheta)]$ intersect O_k . Hence, the ray $\mathcal{R}'_+[h(\vartheta-)]$ is not O_k -free, whereas arbitrarily close to it, there is the upward ray $\mathcal{R}'_+(t)$ that is O_k -free and also lies to the left of $\mathcal{R}$ by Lemma A2 (in the part addressing iv) of Theorem 1). By Definition 2, this means that the ray $\mathcal{R}'_+[h(\vartheta-)]$ is forward critical for O_k , in violation of Assumption 2. This contradiction completes the proof of (i).

(ii) This is immediate from (i) and Definition 3. $\square$

A *right cape* of the obstacle O_k is a pair of its sides s_1, s_2 with a common point p such that they lie to the left of p (see Figure A1).

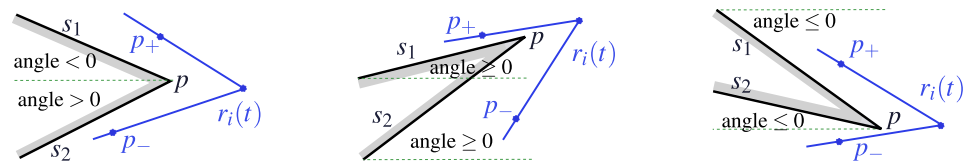


Figure A1. Right capes and feasible options for two non-empty bases.

Lemma A5. *The following statements hold:*

- (i) No more than one obstacle affects the bases $B_i^-(t)$ and $B_i^+(t)$ of USV i .
- (ii) No more than one base is not empty.

Proof. (i) An obstacle influences base building only if vertically below/above its point p there is an edge of the graph $\mathcal{G}_i(t)$ at a distance $\leq \gamma_y$. By the definition of $\mathcal{G}_i(t)$, the y distance from $r_i(t)$ to p does not exceed $\min\{N\gamma_y, R_{\text{vis}}\}$; the x distance does not exceed γ by Corollary A1. By taking into account the imaginary visor, we infer that the x distance to the true obstacle does not exceed $\varkappa + \gamma$. Overall, the Euclidian distance between p and $r_i(t)$ does not exceed g_* , where g_* is the l.h.s. of (20). If building the bases of USV i is affected by different obstacles, the distance between them is, therefore, not greater than $2g_*$, in violation of (20). This completes the proof of (i).

(ii) Suppose that $B_i^+(t) \neq \emptyset$ and $B_i^-(t) \neq \emptyset$, and pick $p_\pm \in B_i^\pm(t)$. Vertically below/above $p^\pm$, there is a point of $O_{k^\pm}$, where $k^+ = k^- =: k$ by (i). Due to Corollary A1 and (17), (1) the x scope of $\mathcal{G}_i(t)$ fully covers the x scope of no side of the polygon O_k , and (2) if $r_i(t)$ lies on the left to a visor of O_k , any node of $\Gamma_i(t)$ to the right of $r_i(t)$ is in the x scope of the visor. Due to (i) in Lemma A4 and the definition of $\mathcal{G}_i(t)$, there exist two sides s_1 and s_2 of O_k such that they form a right cape, and p_+ is vertically above s_1 , whereas p_- is below s_2 ; also, $[p_\pm, r_i(t)] \cap O_k = \emptyset$; see Figure A1.

Based on Figure A1, it is easy to see that, in any case, either (B.3) or (B.2) in the definition of the base (given in Section 5) is violated either for p_+ or for p_- . This contradiction completes the proof. $\square$

The left end of the visor is called its *tip*. The visored obstacle deprived of the visor's tips is said to be *blunt*.

Lemma A6. *The USVs do not hit the blunt obstacles.*

Proof. Let $\mathcal{T}$ be the set of times $t > 0$ when the USVs and the blunt obstacles have no points in common. By Lemma A3, the set $\mathcal{T}$ is non-empty and open, and its leftmost connected component has the form $(0, \tau)$, where $\tau > 0$. We have to show that $\tau = \infty$.

Suppose that $\tau < \infty$ and limit our attention to $t < \tau, t \approx \tau$. With the aid of Lemma A3, we infer that, for such t 's, the USVs are outside the visored obstacles, and there are robot i and obstacle O_k such that $r_i(\tau)$ is in the blunt obstacle O_k , and $r_i(t)$ is either (1) always below O_k or (2) always above O_k . We focus on (1), and case (2) is handled likewise. It is easy to see that there exists a side S of O_k such that $r_i(\tau) \in S$ and $r_i(t)$ is below S ; see Figure A2. Let $p(t)$ stand for the point of S that is vertically above $r_i(t)$. We put $\chi(t) := x[p(t)], y(t) := y[p(t)], z(t) := y(t) - y_i(t)$ and examine the cases from Figure A2 separately.

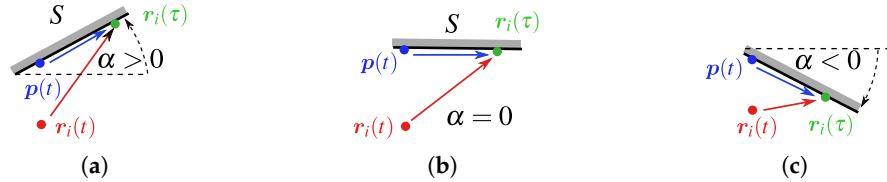


Figure A2. Three ways to hit an obstacle: hitting an obstacle's side with (a) a positive slope, (b) zero slope, (c) negative slope.

The cases (a,b) from Figure A2. Then, $x_i(t) = \chi(t), y(t) = \chi(t) \tan \alpha + \text{const}, d_{i,\delta}^+(t) \leq z(t)$, and due to (4) and Lemma A5, $\alpha_i(t) \leq 0 \forall t < \tau, t \approx \tau$. Hence,

$$\exists \{t_k\}_{k=1}^{\infty} : z(t_k) > 0, t_k < \tau, t_k \rightarrow \tau, z(\tau) = 0; \quad (\text{A2})$$

$$\begin{aligned} \dot{z}(t) &= v_i^x(t) \tan \alpha - v_i^y(t) \stackrel{(5),(7)}{\geq} (v_{\rightarrow} + \bar{F}) \tan \alpha - v_i^y(t) \\ &\stackrel{(8)}{=} (v_{\rightarrow} + \bar{F}) \tan \alpha - G[d_{i,\delta}^+(t)] + \underbrace{G[d_{i,\delta}^-(t)]}_{\geq 0} - \underbrace{P \tan \alpha_i(t)}_{\leq 0} \\ &\geq (v_{\rightarrow} + \bar{F}) \tan \alpha - G[d_{i,\delta}^+(t)] \geq -G[z(t)]. \end{aligned} \quad (\text{A3})$$

Since $z(t_k) > 0$, [65] (Th. 4.1, Ch. 3) guarantees that $z(t)$ is greater than the solution $z_0(t) \equiv 0$ of the ODE $\dot{z}_0(t) = -G[z_0(t)]$ for $t \leq \tau, t \approx \tau$. Then, $z(\tau)$ is also greater than $0 = z_0(\tau)$, in violation of (A2). This contradiction completes the proof in cases (a) and (b).

The case (c) from Figure A2. In this case, robot i is an evader and $\alpha_i(t) = -\alpha_*$ by (4) and Lemma A5. Due to (10) and Lemma A4,

$$\begin{aligned} \dot{z}(t) &= v_i^x(t) \tan \alpha - v_i^y(t) \stackrel{(5),(7)}{\geq} -(v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} - v_i^y(t) \\ &\stackrel{(8)}{=} -(v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} - G[d_{i,\delta}^+(t)] + \underbrace{G[d_{i,\delta}^-(t)]}_{\geq 0} \\ &\quad - P \tan \alpha_i(t) \geq -(v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} - G[d_{i,\delta}^+(t)] + P \tan \alpha_* \stackrel{(15)}{\geq} -G[z(t)] \quad \forall t < \tau, t \approx \tau. \end{aligned}$$

The proof is completed like in cases (a) and (b). $\square$

Corollary A2. *The USVs do not collide with the real obstacles and the corridor walls.*

By the assumptions of Theorem 1, the USVs start outside the caves. Then, (i) in Lemma A4 yields the following.

Corollary A3. *The USVs do not visit the caves.*

By invoking Lemma A3 and A2, we also come to the following two conclusions, respectively.

Corollary A4. Remark 3 is true.

Corollary A5. The times when USV i is in the x scope of any obstacle O_k form a finite interval (maybe, empty). During this interval, USV i is either always above O_k or always below O_k .

For any USVs $i \neq j$, we put $S_{ij}(t) := [r_i(t), r_j(t)]$. A time τ is said to be *touching* for the pair (i, j) and the obstacle O_k if $S_{ij}(\tau) \cap O_k \neq \emptyset$, and arbitrarily close to τ there exists $t < \tau$ such that $S_{ij}(t)$ and O_k have no common points.

Lemma A7. Let τ be a touching time for the pair (i, j) of robots and the obstacle O_k .

Then, the following statements are true:

- (i) Either one and only one of the locations $r_i(\tau), r_j(\tau)$ lies in the x scope of the obstacle O_k , or the both lie there and are vertically aligned $x_i(\tau) = x_j(\tau)$;
- (ii) Any $p \in S_{ij}(\tau) \cap O_k$ is either the left end of a visor or a right cape (see Figure A3e–g);

Proof. We start by noting that the interior of O_k does not intersect with $S_{ij}(\tau)$, and the locations $r_i(\tau)$ and $r_j(\tau)$ do not belong to the blunt obstacle O_k by Lemma A6. So, Corollary A1 and (17) imply that the intersection $S_{ij}(\tau) \cap O_k$ contains a single point p , which is a vertex of the polygon O_k . Let s_1 and s_2 be the sides of O_k adjacent to p . By taking into account (17) and Lemma A2, we see that the cases feasible at $t = \tau$ are either those featured in Figure A3 or their mirror reflections with respect to the horizontal line. We focus on the first group of the cases; the second group is handled likewise.

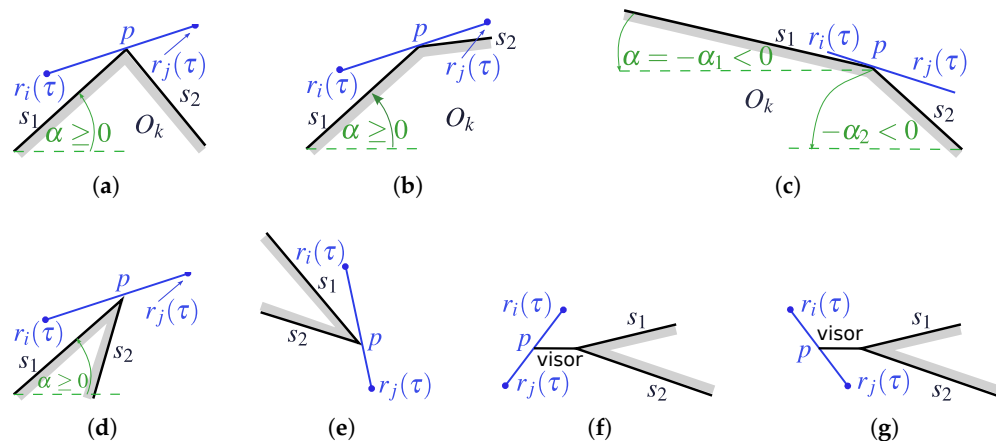


Figure A3. Feasible options at a touching time τ : (a) the x -scopes of the sides s_1 and s_2 do not overlap, the slopes of the sides are opposite in sign, (b) the x -scopes of the sides do not overlap, the slopes of the sides are nonnegative, (c) the x -scopes of the sides do not overlap, the slopes of the sides are negative, (d,e) the x -scopes of the sides overlap, (f,g) the point p is the tip of a visor.

We invoke $\bar{\alpha}$ from (2) and the notation $\varphi(S)$ introduced in the beginning of Section 7, and first focus on the cases (a–d) from Figure A3. By Lemmas A4 and A6 and Figure A3, $|\beta| < \bar{\alpha}$ for $\beta := \varphi[S(\tau)]$. So, Corollary A1 yields that

$$|y_i(\tau) - y_j(\tau)| \leq |x_i(\tau) - x_j(\tau)| \tan |\beta| < \gamma \tan \bar{\alpha} \leq \gamma_y$$

by (17) and the last inequality from (13). Hence, for $t < \tau, t \approx \tau$, nodes i and j are linked in the graphs $\Gamma_i(t)$ and $\Gamma_j(t)$, and $S_{ij}(t) \subset \mathcal{G}_i(t) \cap \mathcal{G}_j(t)$. Also, Lemma A5 and (4) imply that $\alpha_i(t) = \alpha_j(t) = \alpha_*$ in the cases (a), (b), (d), whereas $\alpha_i(t) = \alpha_j(t) = 0$ in the case (c).

In Figure A3, the side s_1 lies on the straight line described by $y = x \tan \alpha + c$ with some c . Also, $p = \theta_i r_i(\tau) + \theta_j r_j(\tau)$, where $\theta_i > 0, \theta_j > 0, \theta_i + \theta_j = 1$. We put $p(t) =$

$\theta_i r_i(t) + \theta_j r_j(t), x(t) := x[p(t)], y(t) := y[p(t)], z(t) := y(t) - x(t) \tan \alpha + c$. Then, (A2) holds and

$$\dot{x} = \theta_i v_i^x + \theta_j v_j^x \begin{cases} \geq v_x \text{ by Lemma A2,} \\ \leq v_{\rightarrow} + \bar{F} \text{ by (5) and (7);} \end{cases}$$

In the cases (a), (b), and (d), we have

$$\begin{aligned} \dot{z} &= \dot{y} - \dot{x} \tan \alpha \geq \dot{y} - (v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} \\ &\stackrel{(8)}{=} -(v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} + \sum_{q=i,j} \theta_q \left\{ \underbrace{G[d_{q,\delta}^+(t)] - G[d_{q,\delta}^-(t)]}_{\geq 0} + P \tan \alpha_* \right\} \\ &\stackrel{(6)}{\geq} -(v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} + \sum_{q=i,j} \theta_q \{-\bar{G} + P \tan \alpha_*\} \xrightarrow{t \rightarrow \tau^-} -(v_{\rightarrow} + \bar{F}) \tan \hat{\alpha} - \bar{G} + P \tan \alpha_* \stackrel{(15)}{>} 0, \end{aligned}$$

in violation of (A2). This shows that (a,b,d) do not occur.

In the case (c) from Figure A3, the sides s_1 and s_2 form a forward pair and so $\bar{\alpha} \geq \alpha_i \geq \beta_{\min}$ by (11) and the definition of $\bar{\alpha}$ from (2). Hence, $|\varphi[S(\tau)]| \leq \bar{\alpha}$, and then (3), (17) and Corollary A1 entail that $d_{q,\delta}^-(\tau^-) \leq \gamma \tan \bar{\alpha}$ for $q = i, j$. Hence,

$$\begin{aligned} \dot{z} &\geq \dot{y} + \dot{x} \tan \alpha_1 \stackrel{\text{Lemma A2}}{\geq} \dot{y} + v_x \tan \beta_{\min} \stackrel{(8)}{=} v_x \tan \beta_{\min} + \sum_{q=i,j} \theta_q \left\{ \underbrace{G[d_{q,\delta}^+(t)] - G[d_{q,\delta}^-(t)]}_{\geq 0} \right\} \\ &\stackrel{(6)}{\geq} v_x \tan \beta_{\min} - \sum_{q=i,j} \theta_q G[d_{q,\delta}^-(t)] \Rightarrow \lim_{t \rightarrow \tau^-} \dot{z}(t) \geq v_x \tan \beta_{\min} - G[\gamma \tan \bar{\alpha}] \stackrel{(18)}{>} 0. \end{aligned}$$

Like in cases (a,b,d), we thus see that (c) does not hold.

In all other cases (e–g) from Figure A3, the statements (i) and (ii) are trivially true. $\square$

Lemma A8. *If an obstacle O_k separates USVs i and j , they cannot be both above O_k and cannot be both below O_k .*

Proof. Suppose to the contrary that there is time t_* such that the obstacle O_k separates $r_i(t_*)$ and $r_j(t_*)$, whereas USVs i and j are either both above O_k or both below O_k . The set $\mathcal{T}$ of times $t \geq 0$ for which O_k does not separate $r_i(t)$ and $r_j(t)$ contains 0 (by the assumptions of Theorem 1), and the time $\tau := \sup\{t \in \mathcal{T} : t < t_*\}$ is touching. Due to Lemma A2 and A7, some of the cases (e–g) from Figure A3 (or its horizontal mirror reflection) hold at $t = \tau$ (maybe, with $i \rightleftharpoons j$). Among them, the case (e) from Figure A3 does not occur: otherwise, the forward USV would not be in the x scope of the obstacle O_k at $t = t_*$ by Lemma A2.

Suppose that $\tau = t_*$. Since the locations $r_i(t_*)$ and $r_j(t_*)$ are in the x scope of O_k , at $t = t_*$, the segment $S_{i,j}(t) := [r_i(t), r_j(t)]$ is vertical in Figure A3f,g. So, one of the USVs i, j is above O_k and the other is below O_k at $t = t_*$, in violation of the definition of t_* . Hence, $\tau < t_*$.

As t runs from τ towards t_* , the forward USV from Figure A3f,g keeps its position with respect to O_k (below/above) unchanged thanks to Corollary A5. Due to Corollary A1 and the inequality $\gamma < \varkappa$ from (17), no later than this USV leaves the x scope of the visor, the other USV enters this scope. Let θ be the time when this occurs. Then, $\tau < \theta \leq t_*$, and at $t = \theta$, the “other” USV is on the other side of the visor since $S_{i,j}(t) \cap O_k \neq \emptyset \forall t \in [\tau, \theta]$. By Corollary A5, it remains on the other side of O_k until t_* , in violation of the definition of t_* . This contradiction completes the proof. $\square$

By the assumption of Theorem 1, the USVs can be enumerated so that, at $t = 0$,

$$y_1(t) < \dots < y_N(t). \quad (\text{A4})$$

Lemma A9. Inequalities (A4) hold for any $t > 0$.

Proof. Suppose that the conclusion of the lemma fails to be true. Let $(0, \tau)$ be the leftmost connected component of the set $\mathcal{T} := \{t > 0 : (\text{A4}) \text{ holds}\}$. Then, $\tau \in (0, \infty)$ and there are $1 \leq j^- < j^+ \leq N$ and $y_{j^*} \in \mathbb{R}$ such that

$$\begin{aligned} y_j(\tau) &= y_{j^*} \quad \forall j \in J := \{j = j^-, \dots, j^+\}, \\ y_{j^- - 1}(\tau) &< y_{j^-}(\tau) \quad \text{if } j^- > 1, \quad y_{j^+}(\tau) < y_{j^+ + 1}(\tau) \quad \text{if } j^+ < N. \end{aligned} \quad (\text{A5})$$

We put $j_{\uparrow} := j^- + 1$, $z(t) := y_{j_{\uparrow}}(t) - y_{j^-}(t)$. By (17) and Corollary A1,

$$|x_{j^-}(t) - x_{j_{\uparrow}}(t)| < \gamma \leq \gamma_x, \quad |y_{j^-}(t) - y_{j_{\uparrow}}(t)| < \gamma_y \quad \text{whenever } t < \tau, t \approx \tau.$$

Also, (A2) still holds. Now, we consider separately several cases.

(1) The segment $S(t) := [r_{j^-}(t), r_{j_{\uparrow}}(t)]$ intersects a blunt obstacle O_k at $t = \tau$. By Lemma A5, the respective obstacle is unique. Among the locations $r_{j^-}(\tau)$ and $r_{j_{\uparrow}}(\tau)$, there is at least one p_{sc} that lies (strictly) inside the x scope of O_k , as follows from (17). We focus on the case where p_{sc} is above O_k ; the opposite case is handled likewise. By Lemma A8, the other p_{sc} of the above two locations is not above O_k . Since the segment $S(\tau)$ is horizontal by (A5) and does not exceed γ in length by Lemma A2, whereas γ is less than the length of the visor by (17), the abscissa $x(p_{\text{sc}})$ cannot lie to the left of the x scope of O_k . Thus, either p_{sc} lies below the obstacle O_k or to the right of its x -scope. We shall handle these two cases separately.

(1.a) Point p_{sc} lies below O_k . By the assumptions of Theorem 1, $S(0) \cap O_k = \emptyset$. Due to Lemma A7 and (e–g) in Figure A3, at the time when the segment $S(t)$ touches O_k for the first time, either USV $j_{\uparrow}$ is above O_k or USV j^- is below O_k . With regard to Corollary A5, it follows that $p_{\text{sc}} = r_{j^-}(\tau)$ lies below O_k and $p_{\text{sc}} = r_{j_{\uparrow}}(\tau)$ lies above O_k . Let s be the side of O_k that contains the point p such that the segment $(p, r_{j_{\uparrow}}(\tau))$ is vertical and does not intersect O_k . Depending on whether the polar angle $\varphi(s)$ of s is non-negative or negative, the situations featured in Figure A4a,b, respectively, occur.

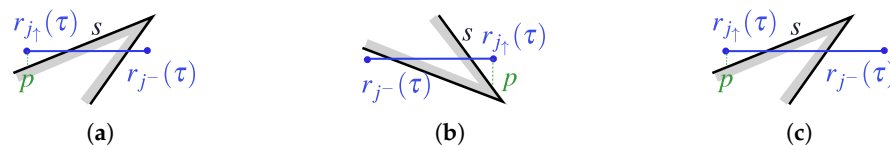


Figure A4. Options feasible at time τ when the ordinates of two USVs become equal: (a) Point p_{sc} lies below the obstacle and $\varphi(s) > 0$, (b) point p_{sc} lies below the obstacle and $\varphi(s) < 0$, (c) point p_{sc} lies to the right of the x scope of the obstacle.

Case from Figure A4a. Like in the proof of Lemma A7, we see that for $t < \tau, t \approx \tau$, USV $j_{\uparrow}$ is an evader and $\alpha_{j_{\uparrow}}(t) = \alpha_*$, $\alpha_{j^-}(t) = 0$. Also, (10) and Corollary A1 imply that $\lim_{t \rightarrow \tau^-} d_{j_{\uparrow}, \delta}^-(t) \leq \gamma \tan \hat{\alpha}$. So for $t < \tau, t \approx \tau$, the following relations hold:

$$\begin{aligned} \dot{z}(t) &= v_{j_{\uparrow}}^y(t) - v_{j^-}^y(t) \stackrel{(8)}{=} G[d_{j_{\uparrow}, \delta}^+(t)] - G[d_{j_{\uparrow}, \delta}^-(t)] - G[d_{j^-, \delta}^+(t)] + G[d_{j^-, \delta}^-(t)] + P \tan \alpha_{j_{\uparrow}}(t) \\ &\stackrel{(6)}{\geq} -G[d_{j_{\uparrow}, \delta}^-(t)] - \bar{G} + P \tan \alpha_* \Rightarrow \lim_{t \rightarrow \tau^-} \dot{z}(t) \geq P \tan \alpha_* - \bar{G} - G[\gamma \tan \hat{\alpha}] \stackrel{(19)}{>} 0, \end{aligned}$$

in violation of (A2). So the case from Figure A4a does not hold.

Case from Figure A4b. Like in the previous case, we see that, for $t < \tau, t \approx \tau$, USV j^- is an evader and $\alpha_{j^-}(t) = -\alpha_*$, whereas $\alpha_{j_\uparrow}(t) = 0$. Also, $\lim_{t \rightarrow \tau^-} d_{j^-, \delta}^+(t) \leq \gamma \tan \hat{\alpha}$. The proof is completed as before and results in the conclusion that the case from Figure A4b does not hold.

Hence, the overall case (1.a) does not occur.

(1.b) Point p_{-sc} lies to the right of the x scope of O_k . Let $\mathcal{T}$ be the set of times $t \in [0, \tau)$ for which the obstacle O_k does not separate USVs i and j . Then, $0 \in \mathcal{T}$ by the assumptions of Theorem 1, the set $\mathcal{T}$ is open in $[0, \tau)$, and $\mathcal{T} \neq [0, \tau)$. So, the leftmost connected component of $\mathcal{T}$ has the form $[0, \theta)$, where $\theta \in (0, \tau)$ is a touching time for (i, j) and O_k . Due to Lemma A2, A7, and (e-g) in Figure A3, at $t \approx \theta, t < \theta$ the USV $j_\uparrow$ with the greater ordinate cannot be below O_k . By Corollary A5 and Lemma A8, it cannot dive below O_k while traveling through the x scope of O_k . Thus, we see that $p_{sc} = r_{j_\uparrow}(\tau)$ and $p_{-sc} = r_{j^-}(\tau)$; see Figure A4c. By arguing like in the situation from Figure A4a, we arrive at the conclusion that the case (1.b) and so the overall case (1) do not occur.

Now, we pass on to consider the remaining option.

(2) Segment $S(\tau)$ does not intersect the blunt obstacles. Since this segment is horizontal, it is also disjoint with the visored obstacles. The USVs j^- and $j_\uparrow$ are in direct sight of each other and

$$S(t) \subset \mathcal{G}_{j^-}(t), \mathcal{G}_{j_\uparrow}(t), \quad d_{j^-, \delta}^-(t) \leq z(t), \quad d_{j_\uparrow, \delta}^+(t) \leq z(t),$$

whereas (A2) holds. Once again, we examine the feasible cases separately.

(2.a) $\alpha_{j^-}(t) \leq \alpha_{j_\uparrow}(t) \forall t < \tau, t \approx \tau$. Then,

$$\begin{aligned} \dot{z}(t) &= v_{j_\uparrow}^y(t) - v_{j^-}^y(t) \stackrel{(8)}{=} \underbrace{G[d_{j_\uparrow, \delta}^+(t)] - G[d_{j_\uparrow, \delta}^-(t)]}_{\geq 0} - \underbrace{G[d_{j^-, \delta}^+(t)] + G[d_{j^-, \delta}^-(t)]}_{\geq 0} \\ &\quad + P \tan \alpha_{j_\uparrow}(t) - P \tan \alpha_{j^-}(t) \stackrel{(6)}{\geq} -2G[z(t)]. \end{aligned}$$

Then, the arguments stated just after (A3) result in contradiction to (A2) and show that the situation (2.a) does not occur.

(2.b) It follows that some of the following options holds:

(b.1) $\tau \in \overline{\mathcal{T}_+}$, where $\mathcal{T}_+ := \{t < \tau : \alpha_{j^-}(t) = \alpha_*, \alpha_{j_\uparrow}(t) \leq 0\}$;

(b.2) $\tau \in \overline{\mathcal{T}_-}$ for $\mathcal{T}_- := \{t < \tau : \alpha_{j^-}(t) \geq 0, \alpha_{j_\uparrow}(t) = -\alpha_*\}$.

(b.1) By (4) and Lemma A5, a unique obstacle affects building any nonzero angle among $\alpha_{j^-}(t), \alpha_{j_\uparrow}(t)$. By shrinking $\mathcal{T}_+$, we arrive at a situation where, for all $t \in \mathcal{T}_+$, the angle $\alpha_{j_\uparrow}(t)$ assumes a common value $\alpha_{j_\uparrow} \leq 0$, a common obstacle O_k influences $\alpha_{j^-}(t)$, and if $\alpha_{j_\uparrow} < 0$, a common obstacle does the same for $\alpha_{j_\uparrow}(t)$. If $\alpha_{j_\uparrow} < 0$, the arguments from the proof of i) in Lemma A5 show that the distance from any USV $j^-, j_\uparrow$ to a point of the related obstacle does not exceed the l.h.s. of (20). Then, (20) implies that both angles are triggered by a common obstacle O_k .

Let $t \in \mathcal{T}_+$. Then, USV j^- is an evader with the main base $B_{j^-}^+(t)$. By (4) with $i := j^-$, there is a node j_* in the connected component $C_{j^-}(t) \ni j^-$ of the graph $\Gamma_{j^-}(t)$ such that $y_{j_*}(t) \leq y_{j^-}(t)$, a side s of O_k can be found vertically below a point $p \in \mathcal{S} := [r_{j^-}(t), r_{j_*}(t)]$ at a distance $\leq \gamma_y$, and the plane-parallel infinitesimal shift of the segment $\mathcal{S}$ and point p in the x direction does not increase the vertical distance from $\mathcal{S}$ to O_k and from p to s . So the polar angle $\varphi(s)$ of the side s is positive. If all nodes from $C_{j^-}(t)$ are in direct view of USV $j_\uparrow$, then $\mathcal{G}_{j^-}(t) \subset \mathcal{G}_{j_\uparrow}(t)$. This and Lemma A5 imply that USV $j_\uparrow$ is also an evader with

the main base $B_{j_{\uparrow}}^+(t)$, in violation of (4) and (b.1). It follows that there is a node $j_{\downarrow} \in C_{j^-}(t)$ invisible by USV $j_{\uparrow}$. Then, necessarily, $y_{j_{\downarrow}}(t) < y_{j^-}(t)$ and so $j_{\downarrow} < j^- < j_{\uparrow}$.

By Corollary A1, the abscissas $x_{j^-}(t)$, $x_{j_{\uparrow}}(t)$, $x(p)$, and $x_{j_{\downarrow}}(t)$ lie in an interval that is less than γ in length. So, (17) implies that $x_{j_{\downarrow}}(t)$ cannot lie on the left to or in the x scope of a visor of O_k . By taking into account the first inequality from (17), we infer that $r_{j_{\downarrow}}(t)$ lies above O_k .

By the assumptions of Theorem 1, O_k does not separate USVs $j_{\uparrow}$ and $j_{\downarrow}$ initially. The time when O_k starts separating them is touching for $(j_{\downarrow}, j_{\uparrow})$ and O_k , and does not exceed t . Let θ stand for the maximum touching time that does not exceed t . By Lemma A7, the situation from Figure A3g,f holds for $\tau := \theta$. In Figure A3g,f, $i = j_{\uparrow}$ and $j = j_{\downarrow}$ so long as $j_{\downarrow} < j_{\uparrow}$ implies that $y_{j_{\downarrow}}(\theta) < y_{j_{\uparrow}}(\theta)$. As time goes from θ to t , the segment bridging the locations of USVs $j_{\uparrow}$ and $j_{\downarrow}$ constantly intersects O_k . Hence, USV $j_{\downarrow}$ is below O_k at the first time when it finds itself in the x scope of O_k . By Corollary A5, USV $j_{\downarrow}$ remains below O_k until t , in violation of the foregoing.

As a result, we see that the case (b.1) does not hold.

(b.2) Similarly, we see that this case also does not occur.

Thus, assuming that the failure of the lemma leads to a contradiction in any case, this completes the proof. $\square$

Now, we introduce $x_{\uparrow} \in \mathbb{R}$ such that the associated concluding part of the corridor $\mathfrak{C}_{+} := \{r \in \mathfrak{C} : x \geq x_{\uparrow}\}$ is free from the obstacles. The foregoing implies the following.

Corollary A6. *The team safely goes through the obstacle course, then enters the concluding part $\mathfrak{C}_{+}$ of the corridor, and goes inside $\mathfrak{C}_{+}$ afterwards.*

From now on, we consider the respective part of the team's motion and assign $t = 0$ to its beginning. We put $y_0(t) := -w/2$, $y_{N+1}(t) := w/2 \forall t \geq 0$, and

$$d_i(t) := y_{i+1}(t) - y_i(t) \quad \forall i = 0, \dots, N. \quad (\text{A6})$$

Lemma A10. *The following function does not decrease:*

$$\mu(t) := \min \left\{ \min_{i=0, \dots, N} d_i(t); \gamma_y \right\}.$$

Proof. By the Danskin theorem [64], the function $\mu(\cdot)$ is absolutely continuous, and for almost all t ,

$$\dot{\mu}(t) = \begin{cases} 0 & \text{if } \min_{i=0, \dots, N} d_i(t) \geq \gamma_y, \\ \dot{d}_{i_t^0}(t) & \text{otherwise,} \end{cases} \quad (\text{A7})$$

where i_t^0 is a minimizer for $\min_i d_i(t)$. It suffices to show that $\dot{\mu}(t) \geq 0$ a.a. t . In the first case from (A7), this is clear. Let the second case hold. We consider three possible options separately.

$i^0(t) = 1, \dots, N-1$. Then, $\gamma_y > \mu(t) = y_{i^0(t)+1}(t) - y_{i^0(t)}(t)$. Since $\sum_{i=0}^N d_i(t) = w$, we infer that $\mu(t) \leq \frac{w}{N+1}$. Hence, $\|r_{i^0(t)+1}(t) - r_{i^0(t)}(t)\| < R_{\text{vis}}$ due to (13), the last inequality from (17), and Corollary A1. So $\mathcal{G}_{i^0(t)}(t) = \mathcal{G}_{i^0(t)+1}(t)$, $\alpha_{i^0(t)}(t) = \alpha_{i^0(t)+1}(t)$, and

$$\begin{aligned} \dot{\mu}(t) &= \dot{d}_{i_t^0}(t) \stackrel{(\text{A6})}{=} \dot{y}_{i^0(t)+1}(t) - \dot{y}_{i^0(t)}(t) \\ &\stackrel{(\text{8})}{=} G[d_{i^0(t)+1, \delta}^+(t)] - G[d_{i^0(t)+1, \delta}^-(t)] - G[d_{i^0(t), \delta}^+(t)] + G[d_{i^0(t), \delta}^-(t)]. \end{aligned} \quad (\text{A8})$$

Here $d_{i^0(t)+1,\delta}^-(t) = d_{i^0(t),\delta}^+(t) = d_{i^0(t)}(t) = \mu(t)$ by (3) since $\|r_{i^0(t)+1}(t) - r_{i^0(t)}(t)\| < R_{\text{vis}}$. Also, $d_{i^0(t)+1,\delta}^+(t) \geq \mu(t)$ and $d_{i^0(t),\delta}^-(t) \geq \mu(t)$ by (3). Overall, $\dot{\mu}(t) \geq 0$ by (6).

$i^0(t) = 0$. Likewise, $\mu(t) = y_1(t) - y_0(t) = |y_1(t) + w/2|$. Due to (12) and (20), the angle $\alpha_1(t)$ cannot be influenced by the upper wall of the corridor since USV 1 is close $|y_1(t) + w/2| < \gamma_y$ to the lower wall. Hence, $\alpha_1(t) \geq 0$ and

$$\dot{\mu}(t) = \dot{d}_0(t) \stackrel{(A6)}{=} \dot{y}_1(t) \stackrel{(8)}{\geq} G[d_{1,\delta}^+(t)] - G[d_{1,\delta}^-(t)] = G[d_{1,\delta}^+(t)] - G[\mu(t)] \geq 0. \quad (A9)$$

$i^0(t) = N$. Similar arguments show that $\dot{\mu}(t) \geq 0$. $\square$

By putting $\zeta := \mu(0) > 0$, we arrive at the following.

Corollary A7. $y_{i+1}(t) - y_i(t) \geq \zeta > 0 \forall t, i$.

Lemma A11. The derivatives $\dot{d}_i(t)$ are bounded: there exists a constant $c > 0$ such that ,

$$|\dot{d}_i(t)| \leq c \quad \text{for any } i = 0, \dots, N \text{ and a.a. } t.$$

Proof. For $i = 1, \dots, N-1$, we have $d_{i+1,\delta}^-(t) = d_{i,\delta}^+(t)$,

$$\begin{aligned} \dot{d}_i(t) &\stackrel{(A6)}{=} \dot{y}_{i+1}(t) - \dot{y}_i(t) \stackrel{(8)}{=} G[d_{i+1,\delta}^+(t)] - G[d_{i+1,\delta}^-(t)] \\ &\quad + P \tan \alpha_{i+1}(t) - G[d_{i,\delta}^+(t)] + G[d_{i,\delta}^-(t)] - P \tan \alpha_i(t) \\ &= G[d_{i+1,\delta}^+(t)] - 2G[d_{i,\delta}^+(t)] + G[d_{i,\delta}^-(t)] + P[\tan \alpha_{i+1}(t) - \tan \alpha_i(t)] \\ &\stackrel{(6),(4)}{\implies} |\dot{d}_i(t)| \leq c := 4\bar{G} + 2P \tan \alpha_*. \quad (A10) \end{aligned}$$

For $i = 0$, we have

$$\dot{d}_i(t) \stackrel{(A6)}{=} \dot{y}_1(t) \stackrel{(8)}{=} G[d_{1,\delta}^+(t)] - G[d_{1,\delta}^-(t)] + P \tan \alpha_1(t) \stackrel{(6),(4)}{\implies} |\dot{d}_i(t)| \leq c. \quad (A11)$$

The case $i = N$ is considered likewise. $\square$

We consider the function $Y(t) := [y_0(t), \dots, y_{N+1}(t)]$ and recall that its ω -limit point is the limit $\lim_{j \rightarrow \infty} Y(t_j)$ born by an unbounded sequence $t_1 < t_2 < \dots < t_j \xrightarrow{j \rightarrow \infty} \infty$. Since the components $y_j(t)$ of $Y(t)$ are bounded $|y_j(t)| \leq w/2 \forall t \geq 0$, the collection $\mathfrak{Y}$ of all ω -limit points is a compact non-empty set. The elements $\mathfrak{y} = [y_{-1}^\infty, \dots, y_N^\infty]$ of this collection are called the *limit deployments*. By Corollary A7, $y_0^\infty = -w/2 < y_1^\infty < \dots < y_{N+1}^\infty = w/2$.

Lemma A12. The following inequality holds for any limit deployment $\mathfrak{y}$:

$$\frac{w}{N+1} \leq d_i^\infty(\mathfrak{y}) := y_{i+1}^\infty - y_i^\infty \quad \forall i = 0, \dots, N. \quad (A12)$$

Proof. First of all, we introduce the continuous mapping

$$\mathfrak{y} \in \mathfrak{Y} \mapsto m(\mathfrak{y}) := \min_{i=0, \dots, N} d_i^\infty(\mathfrak{y}) \in \mathbb{R}. \quad (A13)$$

Suppose to the contrary to the conclusion of the lemma that (A12) fails to be true for some limit deployment $\mathfrak{y}_* \in \mathfrak{Y}$. Then, the minimum of the function $m(\cdot)$ over the compact set $\mathfrak{Y}$

is less than $w/(N+1)$. Let $\bar{y} = [\bar{y}_0^\infty, \dots, \bar{y}_{N+1}^\infty] \in \mathfrak{Y}$ be a respective minimizer. By (A13), $m(\bar{y}) = d_i^\infty(\bar{y})$ for some i , but not for all i 's since

$$\sum_{i=0}^N d_i^\infty(\bar{y}) = w. \quad (\text{A14})$$

By noting that $m(\bar{y}) < w/(N+1)$, we infer that there exists i for which either

$$\frac{w}{N+1} > d_i^\infty(\bar{y}) = m(\bar{y}), \quad m(\bar{y}) < d_{i+1}^\infty(\bar{y}) \quad (\text{A15})$$

$$\text{or} \quad \min\left\{\frac{w}{N+1}, d_i^\infty(\bar{y})\right\} > d_{i+1}^\infty(\bar{y}) = m(\bar{y}). \quad (\text{A16})$$

By definition, the limit deployment $\bar{y}$ is born by some “parent-sequence” $t_1 < t_2 < \dots < t_j \xrightarrow{j \rightarrow \infty} \infty$. By thinning this sequence, if necessary, we ensure that $t_{k+1} \geq t_k + 1 \forall k$. Furthermore,

$$y_j(t_k) \rightarrow \bar{y}_j^\infty \Rightarrow d_j(t_k) \xrightarrow{k \rightarrow \infty} d_j^\infty(\bar{y}) \quad \forall j. \quad (\text{A17})$$

Case (A15). In this case, the following inequality holds if $\kappa > 0$ is small enough

$$m(\bar{y}) < \min\left\{d_{i+1}^\infty(\bar{y}) - 5\kappa; \frac{w}{N+1} - 3\kappa\right\}. \quad (\text{A18})$$

Furthermore, the symbol $\chi(k)$ (maybe, with indices) is used to denote k -dependent quantities that converge to zero as $k \rightarrow \infty$. Borrowing c from Lemma A11, we see that, for $\varepsilon \leq \min\{\kappa/c; 1\}$ and $t \in [t_k - \varepsilon, t_k]$, the following holds:

$$\begin{aligned} d_i(t) &\leq d_i(t_k) + |d_i(t) - d_i(t_k)| \leq d_i(t_k) + c|t - t_k| \leq d_i(t_k) + c\varepsilon \xrightarrow{(\text{A15}), (\text{A17})} m(\bar{y}) + c\varepsilon + \chi_i(k) \\ &\stackrel{k \approx \infty}{\leq} m_* := m(\bar{y}) + 2\kappa < \frac{w}{N+1} - \kappa < \frac{w}{N+1} \xrightarrow{(3), (17), (A4), (A6), \text{Corollary A1}} d_i(t) = d_{i,\delta}^+(t); \\ d_{i+1}(t) &\geq d_{i+1}(t_k) - |d_{i+1}(t) - d_{i+1}(t_k)| \\ &\geq d_{i+1}(t_k) - c|t - t_k| \geq d_{i+1}^\infty(\bar{y}) + \chi_{i+1}(k) - c\varepsilon \\ &\stackrel{k \approx \infty}{\geq} d_{i+1}^\infty(\bar{y}) - 2\kappa \geq m_* + \kappa \xrightarrow{m_* + \kappa < \frac{w}{N+1}, (3), (17), (A4), (A6), \text{Corollary A1}} d_{i+1,\delta}^+(t) \geq m_* + \kappa; \\ j &\leq N-2 \Rightarrow d_j(t) \geq d_j(t_k) - |d_j(t) - d_j(t_k)| \geq \\ d_j(t_k) - c\varepsilon &\xrightarrow{(\text{A17}), d_j^\infty(\bar{y}) \geq m(\bar{y})} \Delta_j(t) \geq m(\bar{y}) + \chi_j(k) - c\varepsilon. \end{aligned}$$

Suppose first that $i \geq 1$. If $\alpha_{i+1}(t) = -\alpha_*$, the graph $\mathcal{G}_{i+1}(t)$ contains a node $j \geq i+1$ at a distance $\leq \gamma_y$ from the upper wall of the corridor. This distance upper estimates $d_j(t)$, where $d_j(t) > d_i(t)$ by the foregoing. Hence, $\Delta_i(t) < \gamma_y$ and so $\mathcal{G}_{i+1}(t) \subset \mathcal{G}_i(t)$

and $\alpha_i(t) = -\alpha_*$. Similarly, $\alpha_i(t) = \alpha_* \Rightarrow \alpha_{i+1}(t) = \alpha_*$. It follows that, in any case, $\alpha_{i+1}(t) \geq \alpha_i(t)$ and so

$$\begin{aligned}
 t \in [t_k - \varepsilon, t_k] &\Rightarrow \dot{d}_i(t) \stackrel{(A10)}{\geq} G[d_{i+1,\delta}^+(t)] - 2G[d_{i,\delta}^+(t)] \\
 &+ G[d_{i,\delta}^-(t)] \stackrel{(6), k \approx \infty}{\geq} G[m(\bar{y}) + 3\kappa] - 2G[m(\bar{y}) + c\varepsilon + \chi_i(k)] \\
 &+ G[d_{i-1,\delta}^+(t)] \geq G[m(\bar{y}) + 3\kappa] - 2G[m(\bar{y}) + c\varepsilon + \chi_i(k)] \\
 &+ G[m(\bar{y}) + \chi_{i-1}(k) - c\varepsilon] \\
 \Rightarrow \lim_{k \rightarrow \infty} \inf_{t \in [t_k - \varepsilon, t_k]} \dot{d}_i(t) &\geq G[m(\bar{y}) + 3\kappa] - 2G[m(\bar{y}) + c\varepsilon] + G[m(\bar{y}) - c\varepsilon] \\
 &\xrightarrow{\varepsilon \rightarrow 0} g := G[m(\bar{y}) + 3\kappa] - G[m(\bar{y})] \stackrel{(6),(13),(A18)}{>} 0.
 \end{aligned}$$

So, by properly reducing ε (if necessary), one can guarantee that $\dot{\Delta}_i(t) \geq g/2$ for all $t \in [t_k - \varepsilon, t_k]$ and large enough k . Therefore,

$$d_i(t_k - \varepsilon) = d_i(t_k) - \int_{t_k - \varepsilon}^{t_k} \dot{d}_i(t) dt \leq d_i(t_k) - g\varepsilon/2. \quad (A19)$$

The bounded sequence $\{Y(t_k - \varepsilon)\}_{k=1}^\infty$ has a convergent subsequence. Its limit y_ε is a limited deployment for which the related quantity (A13) is less than the value taken by this quantity at the deployment $\bar{y}$: this holds due to (A17) (where $t_k \mapsto t_k - \varepsilon$ and $\bar{y} \mapsto y_\varepsilon$) and (A19). However, this contradicts the definition of $\bar{y}$ as the minimizer of (A13). This contradiction completes the proof if $i \geq 1$.

In the case (A15), it remains to examine the situation where $i = 0$. Then, for any $t \in [t_k - \varepsilon, t_k]$, we have

$$\begin{aligned}
 \dot{d}_i(t) &\stackrel{(A11)}{\geq} G[d_{0,\delta}^+(t)] - G[d_{0,\delta}^-(t)] = G[d_{0,\delta}^+(t)] - G[d_{-1,\delta}^+(t)] \\
 &\stackrel{(6), k \approx \infty}{\geq} G[m(\bar{y}) + 3\kappa] - G[m(\bar{y}) + c\varepsilon + \chi_{-1}(k)]
 \end{aligned}$$

Hence,

$$\begin{aligned}
 \lim_{k \rightarrow \infty} \inf_{t \in [t_k - \varepsilon, t_k]} \dot{d}_i(t) &\geq G[m(\bar{y}) + 3\kappa] - G[m(\bar{y}) + c\varepsilon] \\
 &\xrightarrow{\delta \rightarrow 0} g_* := G[m(\bar{y}) + 3\kappa] - G[m(\bar{y})] \stackrel{(6),(13),(A18)}{>} 0.
 \end{aligned}$$

The proof is completed like in the case where $i \geq 1$.

Case (A16) is handled likewise, by systematically replacing the intervals of the form $[t_k - \varepsilon, t_k]$ by $[t_k, t_k + \varepsilon]$ and picking a convergent subsequence from the sequence $\{Y(t_k + \varepsilon)\}_{k=1}^\infty$. $\square$

By using (A12) and (A14), we arrive at the following.

Corollary A8. *The following equality is true for any limit deployment: y ,*

$$d_i^\infty(y) = \frac{w}{N+1} \quad \forall i = 0, \dots, N. \quad (A20)$$

Lemma A13. *For any $i = 1, \dots, N$,*

$$y_i(t) \rightarrow -w/2 + iw/(N+1) \quad \text{as } t \rightarrow \infty. \quad (A21)$$

Proof. By (A12) and (A20), any limit deployment $\mathcal{Y} = [y_0^\infty, \dots, y_{N+1}^\infty]$ satisfies the following equations:

$$y_i^\infty = -w/2 + iw/(N+1) \quad \forall i = 1, \dots, N.$$

Let us consider an ω -limit point $\bar{y}$ of the mapping $y_i(\cdot)$. Then, there exists an unboundedly increasing time-sequence $\{t_k\}$ such that $\bar{y} = \lim_{k \rightarrow \infty} y_i(t_k)$. Furthermore, we note that the bounded sequence $\{Y(t_k)\}_{k=1}^\infty$ has a convergent subsequence. Its limit is a limit deployment whose i th entry is equal to $\bar{y}$. Hence, $\bar{y} = -w/2 + iw/(N+1)$ and we see that all ω -limit points of the bounded function $y_i(\cdot)$ are the same and equal $-w/2 + iw/(N+1)$. This implies (A21) [65] (Ch. VII, Corollary 1.1) and completes the proof. $\square$

Corollary A1, Lemma A13, (13), and the last inequality from (17) imply the following.

Corollary A9. For all sufficiently large t , the graph $\mathcal{G}(t)$ introduced just after (22) is connected; moreover, its nodes i and $i+1$ are connected with an edge for any $i = 1, \dots, N-1$.

Proof of Theorem 1. Corollary A9 and (5), (22) imply that there is x_* for which $z_i(t) \rightarrow x_* \forall i$ as $t \rightarrow \infty$ [59,66]. This, Definition 1, Corollary A2, Lemma A9, and (A21) ensure (ii) in Theorem 1. Its statements (i) and (iii,iv) hold by Lemma A1 and A2, respectively. The statement (v) holds by Lemma A9; the statement (vi) is immediate from (v). $\square$

Remark 3 is justified by Corollary A4.

Proof of Remark 2. The discontinuities could hold either due to the effects of the obstacles or in a situation where $\|r_i - r_j\| = R_{\text{vis}}$ or $|x_i - x_j| = \delta$ for some i, j . However, this cannot occur for all sufficiently large times due to ii) of Theorem 1, Definition 1, the second inequality in (13), Lemma A2, and Corollary A6. $\square$

Proof of Remark 4. This proof is performed via the inspection of the proof of Theorem 1 and observing that all arguments remain valid. Starting from Corollary A6, it is important that a concluding part $\mathcal{C}_+ := \{r \in \mathcal{C} : x \geq x_+\}$ is bounded by two parallel straight lines by (ii) in Definition 5. $\square$

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